Can Plausible Errors Provide Cover for Racial Discrimination?*

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Latest version

Abstract

In this study, I ask whether ambiguous framing can provide moral cover for racial discrimination. The study is motivated by the idea that discriminatory preferences may remain unexpressed when people cannot easily justify acting on them. Ambiguous framing can provide a plausible non-discriminatory explanation for biased behavior. I conducted a preregistered, incentivized experiment with 4,327 White U.S. participants who allocated real monetary bonuses to actual recipients. The experiment varied whether the ambiguously framed payoff option benefited a Black recipient or a White recipient. This design enables me to distinguish between a general tendency to avoid ambiguously framed options and the selective use of ambiguous framing as a justification for discrimination. On average, the results do not support the hypothesis that the framing provides moral cover for discrimination. Participants choose the ambiguously framed option more often when it benefits the Black recipient than when it benefits the White recipient—the opposite of what its selective use against Black recipients would predict. Overall, the evidence is more consistent with a general tendency to avoid ambiguously framed options than with the use of such framing as moral cover for racial discrimination.

Keywords: racial discrimination, motivated reasoning, excuses, online experiment

JEL Classification: C91, D63, D91, J15, D64.

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1 Introduction

Racial discrimination remains economically consequential in the United States[1]. Discrimination has negative consequences across labor markets, housing markets, the justice system, charitable giving, and other domains (Arrow, 1998; Bertrand and Duflo, 2017; Bertrand and Mullainathan, 2004; Edelman et al., 2017; List, 2004; Arnold et al., 2018). Traditional economic models attribute bias to taste-based preferences and/or to (in)accurate beliefs and stereotypes (Becker, 1957; Arrow, 1973; Phelps, 1972; Aigner and Cain, 1977). Nevertheless, in many settings discriminatory preferences remain latent, because acting on them incurs moral, social, or reputational costs (Bénabou and Tirole, 2006; Grossman and van der Weele, 2017)[2]. Therefore, it is a fundamentally important challenge to both pinpoint where such preferences exist and to understand the contexts that unlock them.

This paper investigates whether ambiguous framing in the decision environment can activate latent racial bias by providing a "plausible excuse" for discriminatory behavior.[3] I characterize the selective use of the framing as a *motivated interpretation* and the resulting behavior as *excuse-driven discrimination*. This mechanism is relevant to many consequential economic decisions. Job postings may contain loosely defined requirements, performance evaluations often rely on subjective judgments, and grading practices may leave room for discretion. In each setting, criteria that are indirect or open to interpretation may provide seemingly non-discriminatory explanations for choices influenced by racial preferences, making discrimination harder to identify.

To examine the role of ambiguous framing as a cover for racial discrimination, I conduct a pre-registered, incentivized experiment in which participants allocate monetary bonuses across three conditions. First, in the *Direct Black–White (DirectBW)* task, participants choose between a sure 100-cent bonus payment for a Black profile and a sure 100-cent bonus payment for a White profile. The two options are payoff-equivalent, and this task provides a standard experimental measure of discriminatory behavior (Fershtman and Gneezy, 2001; List, 2004; Lane, 2016; Enke et al., 2021). However, this transparent setting

[1]For example, the U.S. Bureau of Labor Statistics reports that median weekly earnings for Black full-time workers is $970, and $1,238 for White workers, a gap of 21.7% (U.S. Bureau of Labor Statistics, 2025). While economic disparities arise via multiple channels, a large body of research shows that racial discrimination is an important contributor to these gaps (Derenoncourt et al., 2024; DeSilva and Elmelech, 2012).

[2]Bénabou and Tirole (2006) theoretically show how concern for social reputation and self-image can discipline behavior by making undesirable motives costly to reveal. Grossman and van der Weele (2017) combine theory and experimental evidence to show that uncertainty and/or ignorance can mitigate those image costs by providing an excuse for self-serving behavior.

[3]I use *ambiguous framing* in its ordinary sense to describe a payoff stated in a less direct and potentially confusing way. It does not refer to ambiguity in the economic sense of unknown probabilities (Ellsberg, 1961). The study does not treat ambiguous framing as a distinct source of discrimination; rather, it examines whether taste-based or stereotype-based discrimination is more likely to be expressed when the decision environment provides a plausible excuse for it.

may make it difficult to justify a choice against the Black recipient on grounds other than race. As a result, participants with latent discriminatory preferences may refrain from acting on them even when/if they in fact hold such preferences. To make this behavior psychologically easier, I then introduce the *Indirect Black–White (IndirectBW)* task. Here, the payoff option attached to the Black recipient is described as "the larger amount between 100 and 50 cents" (which is always 100 cents), while the White-recipient option remains a sure 100 cents. The actual payoff remains the same, but the ambiguous wording allows a potential justification: participants can attribute their choice to the description rather than to race. Finally, in the *Indirect White–White (IndirectWW)* task, both recipients are White and the same ambiguous framing is attached to one of them. This allows me to isolate the effect of the framing from racial bias.

I test two pre-registered hypotheses. First, I compare DirectBW and IndirectBW to examine whether ambiguous payoff description reduces the probability of choosing the Black recipient. If participants select the Black profile more often in DirectBW than in IndirectBW, this provides initial evidence that ambiguous framing enables discrimination. However, because this reduction of selecting the Black profile may reflect a general aversion to indirect descriptions, I test a second hypothesis by comparing IndirectBW and IndirectWW. These tasks are framed in equally ambiguous ways, with the ambiguous option attached to a Black recipient in IndirectBW and a White recipient in IndirectWW. If participants systematically use indirect framing as a cover for discrimination, they should avoid that option more when it benefits the Black recipient than when it benefits the White recipient.

The results are as follows. I find no baseline discrimination in the data: participants choose the Black recipient in 52.6% of cases in DirectBW, which suggests no substantial racial bias. In line with the first hypothesis, when the ambiguous framing is attached to the Black recipient in IndirectBW — and the White-recipient option has the sure amount — the probability of choosing the Black recipient falls by 7 percentage points. This shift alone is consistent with the idea that ambiguous framing facilitates discrimination. However, the test of my second hypothesis — the more informative comparison across equally ambiguous tasks — fails to support a motivated interpretation of the framing to discriminate on average in the experimental sample. Participants choose the ambiguously framed option in 40.1% of cases when it is attached to the White recipient and in 45.6% of cases when it is attached to the Black recipient. Thus, participants are more likely to choose the ambiguously framed option when it benefits the Black profile than when it benefits the White profile. This pattern contradicts the moral cover hypothesis: participants are not avoiding the ambiguous option because it benefits the Black recipient. On average in the experimental sample, the results are more consistent with a general tendency to avoid ambiguously framed options than with their selective use as a cover for discrimination

against Black recipients[4].

As a next step, I examine whether the average pattern masks meaningful heterogeneity across subgroups. I examine whether patterns consistent with excuse-driven discrimination emerge among respondents with more negative racial attitudes toward Black Americans. This is motivated by research suggesting that contextual factors may be particularly significant for individuals who already hold negative stereotypes, as permissive environments can make it easier to justify the expression of those attitudes (Newman et al., 2021). To examine this possibility, I construct a *Composite Racial Attitudes Index* that combines six pre-registered items measuring attitudes toward Black Americans. Two items ask respondents to evaluate Black Americans based on traits related to work ethic and intelligence. Four additional items ask respondents to express their agreement with statements concerning prejudice, discrimination, and individual effort. Higher levels of the Composite Racial Attitudes Index indicate stronger negative attitudes towards Blacks.

Heterogeneity analysis using this composite measure shows substantial baseline heterogeneity. In DirectBW, the share choosing the Black recipient is 71.5% in the Low composite-attitudes group, 49.2% in the Medium group, and 36.9% in the High group. Attaching the ambiguous framing to the Black-recipient option reduces these shares to 60.9%, 43.9%, and 31.7%, respectively. The reductions are statistically significant in all three groups ($p < 0.001$, $p = 0.004$, and $p = 0.001$). Although the absolute reductions differ, relative to each group's DirectBW baseline they do not follow a clear downward pattern: 14.9%, 10.6%, and 14.0%. In fact, the proportional reductions in the Low and High groups are similar. The smaller absolute reduction in the High group therefore largely reflects its lower starting probability of choosing the Black recipient.

The comparison between IndirectBW and IndirectWW is more directly informative about whether choices depend on the race of the recipient who benefits from ambiguous framing. In the Low composite-attitudes group, 60.9% choose the ambiguous option when it benefits the Black recipient, compared with 40.0% when it benefits the White recipient ($p < 0.001$). This pattern reverses in the High group: 31.71% choose the ambiguous option when it benefits the Black recipient, compared with 39.36% when it benefits the White recipient ($p < 0.001$). At the level of group means, the High-group reversal moves in the direction predicted by the moral-cover explanation.

However, an exploratory analysis of individual choices weakens this interpretation. The High-group reversal is driven by participants who already choose the White recipient in DirectBW, while the choices of participants who select the Black recipient in DirectBW

[4]The recipient profiles represent real people and, in addition to race, display sport, color, and drink preferences. These attributes are not balanced by race in the recipient pool. Appendix D documents these differences, and Section 3.3 examines the sensitivity of the results to profile composition.

do not distinguish moral cover from a more general tendency to avoid ambiguous options (see Section 3.4).

This paper contributes to two strands of literature. First, it contributes to the literature on racial discrimination by examining the conditions under which latent discriminatory preferences manifest in observable behavior. Bauer and Chytilová (2025) discuss how discriminatory preferences are more likely to manifest when the environment weakens social norms. Similarly, Bauer et al. (2018) show that ethnic hostility can emerge when individuals observe hostile peer behavior. Research also shows that political signals — such as electoral results or local political support — can similarly relax constraints on openly expressing intolerance (Bursztyn et al., 2020; Edwards and Rushin, 2018; Newman et al., 2021; Grosjean et al., 2023). Related work finds that harmful behavior toward minorities is more readily expressed when it can be framed as a punishment or when the allocation of responsibility is unclear (Bauer et al., 2023, 2024). I study a related mechanism: whether ambiguous payoff description makes a discriminatory choice easier to rationalize.

Second, this study contributes to the literature on motivated errors by testing whether ambiguous framing can create moral wiggle room and serve as a cover for acting on intrinsic discriminatory biases. People often rationalize their actions by exploiting uncertainty, attributing choices to benign preferences or naive mistakes (Dana et al., 2007; Spiekermann and Weiss, 2016; Exley, 2016; Exley and Kessler, 2024; Haisley and Weber, 2010). This reasoning is closely related to motivated reasoning and self-image concerns (Kunda, 1990; Epley and Gilovich, 2016; Bénabou and Tirole, 2006; Grossman and van der Weele, 2017). In particular, Exley and Kessler (2024) show that individuals are more likely to make "errors" when the errors help to justify selfish behavior. In relation to discrimination, Eyting (2026) shows that individuals search for and interpret information about workers from different racial and ethnic groups selectively, forming directional beliefs that, in turn, lead to discriminatory hiring decisions. Similarly, ambiguous framing can be interpreted in self-serving ways, especially when it rationalizes unfair other-regarding behavior (Haisley and Weber, 2010). Rather than focusing on belief formation, I ask whether ambiguous framing selectively affects how pre-existing discriminatory preferences translate into observed choices. In the context of discrimination, the key prediction is not merely that ambiguous framing changes choices, but that it does so selectively when it is attached to a minority recipient (Barron et al., 2025).

The paper proceeds as follows. In Section 2, I describe the experimental design, sample, and measures. In Section 3, I present the main results, robustness of the findings, and discuss heterogeneity in treatment effects. Section 4 concludes.

2 Experimental Design

The experiment has two stages and was implemented on Prolific, which has been suggested to provide higher data quality than similar platforms (Peer et al., 2022)[5]. In Stage 1, I recruited 50 US-based participants and collected data to construct the recipient profiles for Stage 2. The primary purpose of Stage 1 was to create a pool of real recipients with standardized profiles, so that Stage 2 decisions correspond to allocations to real individuals. Each profile includes a race cue and a small set of payoff-irrelevant attributes — favorite color, sport, tea vs. coffee preference[6]. Using profiles of actual Stage 1 participants makes allocation choices consequential for real recipients. The trade-off is that their displayed sport, color, and drink preferences are not balanced across races. Within each task, two distinct profiles are randomly drawn from the relevant recipient pools. Draws are made separately across tasks, so a profile may appear again in another task, potentially paired with a different recipient. To account for observed profile differences, I include profile controls in the main regressions and examine sensitivity to profile composition in Section 3.3 and Appendix D[7].

In Stage 2, a separate sample of participants completed a sequence of incentivized allocation tasks. In each task, the participant chose between two recipient profiles, and their choice determined the recipient's bonus payment (either 100 or 0 cents). Participants were informed that choices are payoff-relevant for real recipients from Stage 1 and that one of their tasks would be randomly selected for payment at the end of the study[8].

The main sample consists of US residents recruited on Prolific using pre-screening. To keep the test of discrimination against Black recipients clean and interpretable, I restrict the participant pool to individuals who self-identify as White. This approach reduces confounding from own-group identity and allows the race cue manipulation to be interpreted as out-group allocations. The main sample includes $4,327$ participants aged from 18 to 64 (mean age 40.3 years). The sample is 53.8% female; 39.8% report being conservative or very conservative, and 38.2% report being liberal or very liberal. At least 65.2% have a Bachelor's degree, most are employed (82.9%), and 61.9% identify as Christian. Table 1

[5]Full experimental instructions are available in Appendix I.

[6]Following Bohren et al. (2025), the profiles include neutral, payoff-irrelevant details so that race is not the only salient characteristic. The purpose is to place the race cue in a richer profile environment, closer to real-world evaluations.

[7]The controls capture whether the Black recipient, relative to the White recipient, prefers tea and shares the participant's sport, color, and drink preferences, as well as whether the Black recipient appears on the left. For the IndirectBW–IndirectWW comparison, analogous controls are defined for the ambiguously framed option relative to the certain option.

[8]Specifically, participants were informed that at the end of the experiment, one task would be randomly chosen; the option chosen in that task would be implemented, and the corresponding Stage 1 recipient would receive the implied bonus. Participants in Stage 1 received their Prolific participation payment in addition to any bonus earnings from the Stage 2 tasks.

reports summary statistics for the main outcomes, heterogeneity measures, and baseline covariates used in regressions; Appendix Table A.1 reports full descriptive statistics.

Each participant made allocation choices under both the simple and ambiguous payoff framings, which allows for comparisons within the same individual. There are four tasks that vary along two dimensions: i) whether the task has a Black versus a White recipient or two White recipients, and ii) whether one option contains an as-if ambiguous payoff structure (see Figure B.1 for illustration of the tasks). In addition, participants were assigned to either a male-profile version or a female-profile version of the task battery, and completed all four tasks using recipient profiles of that assigned gender. The order of tasks is randomized at the participant level, and within each task, the position of profiles into left and right is also randomized. I used the tasks to measure baseline racial discrimination, identify the effect of ambiguous framing when attached to the Black recipient, and examine whether this effect depends on the race of the recipient who benefits from the framing, rather than reflecting a general tendency to avoid ambiguously framed options. The tasks are presented as follows:

1. Simple Black–White (**DirectBW**, baseline): 100 cents for a Black profile vs. 100 cents for a White profile (no ambiguous framing). This task provides a clean baseline for racial discrimination.
2. Ambiguous Black–White (**IndirectBW**, ambiguous framing attached to Black): the larger amount between 100 and 50 cents for a Black profile vs. 100 cents for a White profile. This task introduces an as-if ambiguous payoff structure attached to the Black option.
3. Ambiguous White–White (**IndirectWW**, ambiguous framing attached to White): the larger amount between 100 and 50 cents for a White profile vs. 100 cents for a White profile. This task matches IndirectBW in wording and payoff structure but removes the race cue.
4. Simple White–White (**DirectWW**, placebo): 100 cents for a White profile vs. 100 cents for a White profile[9].

The condition labels describe the racial composition of the recipient pair (B = Black, W = White), not the experimental treatment.[10] The experimental variation is in the payoff framing: whether an option is described ambiguously or with certainty, and which recipient-pair design this ambiguous framing appears in.

[9]DirectWW is a race-neutral placebo. It serves as a diagnostic benchmark for generic left-right choice patterns and as a neutral baseline in the task duration analysis. Appendix E presents these results.

[10]The preregistration uses the labels SimpleBW, AmbigBW, AmbigWW, and SimpleWW for the conditions called DirectBW, IndirectBW, IndirectWW, and DirectWW in the manuscript.

All four tasks are constructed so that the payoff is fixed at 100 cents while the framing varies. In the ambiguous tasks, the Black- (or White-) recipient option is described as "the larger amount between 100 and 50 cents." Since the maximum of {100, 50} is always 100, this option is payoff-equivalent to the sure 100-cent option. The purpose of this design is not to change incentives or information, but to introduce an as-if ambiguous description that can be interpreted (or portrayed) as involving uncertainty or confusion[11]. This creates scope for the ambiguous framing to serve as a justification device — participants can attribute unequal treatment to the framing rather than to race — while keeping the allocation choice problem identical in expected payoffs.

The key comparison to isolate the strategic use of the ambiguous framing is IndirectBW vs. IndirectWW, which holds the payoff wording fixed and compares choices when the ambiguous wording is attached to a Black or a White recipient. This comparison aims to separate a general tendency to avoid ambiguously framed options from discrimination that specifically exploits the framing as a justification device against Black recipients.

The two primary outcomes are: (i) **black chosen**, an indicator equal to 1 if the Black profile is selected in DirectBW and IndirectBW; and (ii) **ambiguous chosen**, an indicator equal to 1 if the ambiguous option is selected in IndirectBW and IndirectWW. Each outcome is analyzed in a separate regression that pools observations from the two relevant tasks only: **black chosen** is estimated on the DirectBW and IndirectBW sample, and **ambiguous chosen** on the IndirectBW and IndirectWW sample. Using these main outcomes, I test two pre-registered hypotheses. First, if ambiguous framing provides cover for discriminatory behavior, participants should be less likely to choose the Black recipient in IndirectBW than in DirectBW (H1). Second, holding the payoff wording fixed, participants should be less likely to choose the ambiguous option in IndirectBW than in IndirectWW (H2).

If excuse-driven discrimination is present, it is more likely to appear among participants who hold more negative racial attitudes toward Black Americans. Therefore, in addition to the main outcomes, I elicit six pre-registered self-reported attitude items toward Black Americans. Two items are stereotype trait ratings adapted from Katz and Braly (1933): participants rate Black Americans on 1–7 scales ranging from hard-working to lazy and from intelligent to unintelligent. Four additional items form a symbolic-racism battery based on Henry and Sears (2002). These items capture agreement with statements emphasizing individual effort by Blacks and disagreement with statements recognizing the lasting consequences of slavery and discrimination or the possibility that Blacks generally receive less than they deserve. Item wording is reported in Appendix G.

[11]Because the ambiguous wording could be misread as implying lower expected payoffs, I include a post-task clarity question and show that the main results are robust when they are restricted to participants who report the wording was clear (see Section 3.3 for more details).

I combine these six items into a *Composite Racial Attitudes Index*. First, I transform all items so that higher values indicate more negative attitudes toward Black Americans. Because the stereotype items are measured on 1–7 scales and the symbolic-racism items on 1–5 scales, I rescale each item to the unit interval and then take their simple average. The resulting raw index ranges from 0 to 1, with higher values indicating more negative racial attitudes. Appendix G.1 shows its distribution. In the regression analysis, I use the raw continuous index. For graphical presentation and descriptive comparisons, I divide the raw index into Low, Medium, and High groups using sample tertiles, so the cutoffs are determined by the empirical distribution of the index. Because these attitude measures were collected after the main tasks, I treat the heterogeneity analysis as exploratory. Appendix G additionally reports robustness analyses using the two underlying components separately: the Stereotype Index (2 items) and the Symbolic Racism Index (4 items).

3 Results

3.1 Does ambiguous framing change allocation behavior?

I start by analyzing whether ambiguous framing changes participants' behavior in allocation decisions between Black and White recipients. In the simple baseline task (DirectBW), participants choose between allocating the same sure amount to a Black recipient or to a White recipient. I find that 52.6% of participants choose the Black profile in DirectBW (Figure 1, Panel A), indicating no substantial baseline discrimination against the Black recipient in the experimental sample.

Introducing the ambiguously framed payoff for the Black-recipient option — while keeping the White-recipient option at the same sure amount as in DirectBW — reduces the share that selects the Black profile from 52.6% in DirectBW to 45.6% in IndirectBW (Figure 1). This initial comparison between DirectBW and IndirectBW is consistent with the idea that ambiguous framing enables discrimination. Table 2 column 1 formalizes these results, showing that introducing ambiguous framing in IndirectBW reduces the frequency of choosing the Black recipient by 7 percentage points. The estimate is statistically significant and remains essentially unchanged when I add the full set of controls: the coefficient moves only slightly, from -0.070 in column 1 to -0.073 in column 2. Because the within-subject specifications include participant fixed effects, the effect is identified from changes in the same individual's choices across tasks and is therefore not driven by time-invariant factors.

The negative effect also appears in the first-task-only between-subject analysis, although it is smaller in magnitude (-0.033 to -0.038) and much less precisely estimated. Thus, the results show that introducing ambiguous framing reduces the choice of the Black profile

Table 1: Summary statistics ($N = 4,327$)

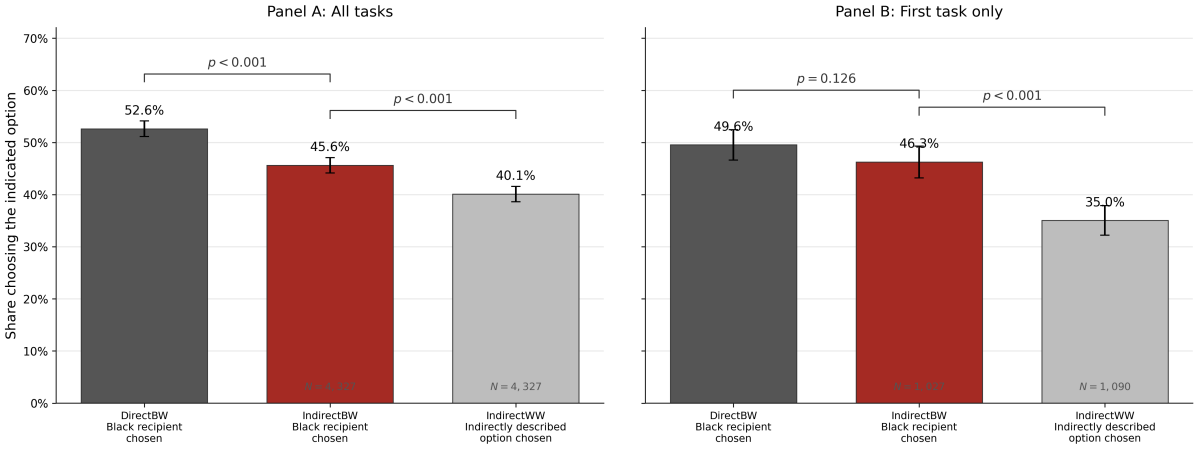
Variable	Definition	Mean	SD	Range
Panel A: Outcomes				
DirectBW	=1 if the Black-recipient option is selected	0.526	0.499	$[0, 1]$
IndirectBW	=1 if the Black-recipient ambiguous option is selected	0.456	0.498	$[0, 1]$
IndirectWW	=1 if the ambiguously framed option is selected	0.401	0.490	$[0, 1]$
Panel B: Controls (compact)				
Age	Years	40.248	11.291	$[18, 64]$
Female	=1 if female	0.538	0.499	$[0, 1]$
BA+	=1 if Bachelor's degree or higher	0.653	0.476	$[0, 1]$
Christian	=1 if Christian	0.619	0.486	$[0, 1]$
Liberal	=1 if Liberal or Very Liberal	0.382	0.486	$[0, 1]$
Conservative	=1 if Conservative or Very Conservative	0.398	0.490	$[0, 1]$
Above median income	=1 if income is at or above the sample median category	0.521	0.500	$[0, 1]$
Panel C: Heterogeneity measures				
Composite Racial Attitudes Index	Average of six racial-attitude items; higher values indicate more negative attitudes toward Black Americans; raw index on $[0, 1]$	0.375	0.218	$[0, 1]$
Stereotype Index	Mean of two trait-rating items; 1–7 scale, in 0.5-point increments; higher values indicate more negative stereotypes	2.695	1.483	$[1, 7]$
Symbolic Racism Index	Mean of four items; 1–5 scale, in 0.25-point increments; higher values indicate more racial resentment	2.684	1.042	$[1, 5]$

Notes: Panel A reports participant × condition means. In DirectBW and IndirectBW, the outcome equals 1 if the participant selects the Black recipient option. In IndirectBW and IndirectWW, the outcome equals 1 if the participant selects the ambiguously framed option. Regression controls additionally include the full set of category indicators for education, ideology, religion, and income. For compactness, the table reports grouped indicators only. BA+ denotes Bachelor's degree or higher. Liberal and Conservative pool (Very) Liberal and (Very) Conservative, respectively. Above median income is defined as income at or above the sample median category ($75–99k).

in a statistically robust way.

This reduction can be interpreted in two ways. One explanation is that participants with discriminatory preferences exploit the ambiguous framing to justify discriminating against the Black recipient because the framing helps to reduce their internal discomfort or feelings of explicit responsibility. This is the excuse-driven discrimination explanation of the findings. Another possibility is that participants avoid the ambiguously framed option regardless of which recipient it is attached to, mechanically reducing Black choice when ambiguous framing is paired with the Black recipient, i.e., there is general tendency to avoid ambiguously framed options. The next step explores these two possibilities.

Figure 1: Mean allocation choices under race and ambiguous framing



Notes: For DirectBW and IndirectBW, the outcome is whether the Black-recipient option is selected. For IndirectWW, the outcome is whether the indirectly described option is selected. Panel A reports mean choices across all tasks, with each participant contributing one observation to each task. Panel B reports mean choices among participants who encountered the relevant task first. Error bars show 95% confidence intervals.

Table 2: Effects of race and ambiguous framing on allocation decisions

	Within-subject				Between-subject (first task only)			
Sample:	*DirectBW + IndirectBW*		*IndirectBW + IndirectWW*		*DirectBW + IndirectBW*		*IndirectBW + IndirectWW*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
IndirectBW	-0.070***	-0.073***	0.055***	0.070***	-0.033	-0.038*	0.112***	0.122***
	(0.010)	(0.010)	(0.010)	(0.010)	(0.022)	(0.021)	(0.021)	(0.022)
Control mean	0.526	0.526	0.401	0.401	0.496	0.496	0.350	0.350
Participant fixed effects	Yes	Yes	Yes	Yes	–	–	–	–
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	8,654	8,654	8,654	8,654	2,145	2,145	2,117	2,117
Clusters (participants)	4,327	4,327	4,327	4,327	–	–	–	–

Notes: Each regression pools observations from two tasks only. The dependent variable is an indicator equal to 1 if the Black-recipient option is selected (columns (1)–(2) and (5)–(6); sample: DirectBW and IndirectBW), or equal to 1 if the ambiguously framed option is selected (columns (3)–(4) and (7)–(8); sample: IndirectBW and IndirectWW, where the ambiguous option is attached to the Black recipient in IndirectBW and to the White recipient in IndirectWW). The treatment indicator IndirectBW equals 1 for IndirectBW observations and 0 for the control-condition observations (DirectBW in columns (1)–(2) and (5)–(6); IndirectWW in columns (3)–(4) and (7)–(8)). "Control mean" is the mean of the dependent variable in the control condition. Within-subject specifications include participant fixed effects and standard errors clustered by participant. First-task specifications use heteroskedasticity-robust standard errors. Controls in columns (2) and (4) additionally include variables for payoff-irrelevant profile attributes. In the DirectBW/IndirectBW comparison, these controls indicate whether the Black profile rather than the White profile has tea as the stated drink preference, whether it shares the participant's stated preferences for sport, color, or drink, and whether it appears on the left side. In the IndirectBW/IndirectWW comparison, analogous controls are defined for the ambiguously framed option relative to the certain option. Controls in columns (6) and (8) include age, gender, education, income, political ideology, and religion, and, additionally, the same profile-composition and position controls used in columns (2) and (4). $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.1$.

3.2 Excuse-driven discrimination or general avoidance of ambiguous framing?

To distinguish between these explanations, I compare two equally ambiguous tasks — IndirectBW and IndirectWW. In IndirectBW, the ambiguously framed option is attached to the Black recipient, whereas in IndirectWW, it is attached to the White recipient. These tasks hold the payoff wording fixed, and the outcome variable is an indicator equal to 1 if the participant selects the ambiguously framed option.

If ambiguous framing provides moral cover for discrimination, then participants should be relatively more willing to choose the ambiguously framed option when it is attached to the White profile than to the Black profile. This implies a higher choice rate of the ambiguously framed option in IndirectWW than in IndirectBW. Figure 1 shows the opposite pattern: the ambiguously framed option is chosen 40.1% of the time in IndirectWW and 45.6% of the time in IndirectBW ($p < 0.001$). In other words, when the ambiguously framed option is attached to the Black recipient, participants choose it 5.5 percentage points more often than when it is attached to the White recipient. Table 2 column 3 formalizes this result. These findings suggest that the results are driven by general tendency to avoid ambiguously framed options rather than by a strategic use of ambiguous framing as moral cover to discriminate against the Black recipient in the experimental sample. The results are robust to alternative specifications and tests (see Section 3.3).

3.3 Robustness

In this section, I present several robustness checks. First, I re-estimate the main treatment effects using only the first tasks for each participant, which addresses concerns specific to within-subject designs. To further support the validity of the within-subject analysis, I test its main assumptions — differential attrition, temporal instability, and carryover effects across conditions. Next, I show that the results are robust to an alternative conditional logit specification with participant fixed effects. I also examine whether the findings are sensitive to low-engagement responding by excluding exceptionally fast participants, and to possible misunderstanding of the ambiguous wording by restricting the sample to participants who report that the framing was clear. Finally, I test whether the results could be driven by payoff-irrelevant drink composition of the profiles by focusing on choices in which the options for comparison share the same stated drink preference. As an additional check, Appendix H shows that the main qualitative pattern is similar across the male-profile and female-profile versions of the tasks.

Between-subject evidence from first-task choices

I first re-estimate the main effects using only the first tasks of each participant. Restricting the sample to the first-task choices provides a between-subject benchmark and addresses concerns that are specific to within-subject designs, such as demand effects, learning, and order-related concerns. Table 2 columns (5)–(8) and Figure 1, Panel B, present the effect of introducing ambiguous framing on participants' allocation decisions using between-subject variation from each participant's first task only. In the baseline DirectBW task, the probability of choosing the Black recipient is 49.6%, close enough to 50% to suggest that there is no baseline discrimination in this subsample.

Introducing ambiguous framing reduces the allocation choices to the Black recipient by 3.5 percentage points. In a comparison between two equally ambiguous tasks, participants choose the ambiguously framed option 11.3 percentage points more often when it is attached to the Black-recipient option than when it is attached to the White-recipient option. Although these results are statistically weaker, consistent with lower power in the between-subject sample, they point in the same direction as the main (within-subject) results.

Within-subject validity

The main analysis exploits within-subject variation in the data: each participant makes allocation choices under both the simple and the ambiguous payoff framings. One advantage of a within-subject design is that it increases precision by controlling for stable individual differences. However, this design can be affected by issues such as selective attrition across tasks, order effects, and carryover or learning effects, where exposure to one task influences behavior in subsequent tasks.

All participants in the sample complete all four tasks, and there is no evidence of differential attrition across conditions. Next, the treatment effect is temporally stable over the task sequence. I test whether the effect of attaching ambiguous framing to the Black-recipient option (IndirectBW) depends on its position in the sequence by interacting the IndirectBW indicator with task-order dummies. The order interactions are jointly insignificant in both comparisons (Appendix Table F.1), indicating that the estimated treatment effects do not systematically vary by task position. Finally, there is no evidence of carryover (causal transience). Sequence tests examine whether the treatment effect depends on whether IndirectBW is seen before or after the corresponding comparison condition. The sequence indicators are small and statistically insignificant, which suggests that exposure to one condition does not systematically change behavior in the other. Appendix F reports formal checks of the main assumptions that within-subject analysis requires in order to preserve internal validity.

Taken together, these checks support that the within-subject estimates are internally valid. I therefore present within-subject results first, and show that the same qualitative results appear in the first-task (between-subject) analysis as a robustness check.

Alternative specification

As an additional robustness check, I estimate conditional logit models with participant fixed effects. Table 3 shows that the qualitative conclusions are unchanged. In the DirectBW–IndirectBW comparison, attaching ambiguous framing to the Black-recipient option is associated with 29–32% lower odds of choosing the Black recipient. In the IndirectWW–IndirectBW comparison, the odds of choosing the ambiguously framed option are 31–46% higher when that option benefits the Black recipient rather than the White recipient. Thus, the main conclusions do not depend on the linear probability specification.

Table 3: Effects of race and ambiguous framing on allocation decisions: conditional logit estimates

Sample:	*DirectBW + IndirectBW*		*IndirectBW + IndirectWW*	
	(1)	(2)	(3)	(4)
IndirectBW (log-odds)	-0.343***	-0.388***	0.268***	0.375***
	(0.048)	(0.050)	(0.048)	(0.056)
Odds ratio exp(β)	0.709	0.678	1.307	1.455
95% CI for OR	[0.646, 0.779]	[0.615, 0.749]	[1.190, 1.435]	[1.302, 1.624]
Control mean	0.585	0.585	0.434	0.434
Participant fixed effects	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Observations	3,576	3,576	3,594	3,594
Clusters (participants)	1,788	1,788	1,797	1,797

Notes: Each regression pools observations from two tasks only. Conditional logit with participant fixed effects estimated on balanced within-subject samples. Columns (1)–(2) use the DirectBW and IndirectBW sample; columns (3)–(4) use the IndirectBW and IndirectWW sample. Participants with no within-person variation in the dependent variable are dropped by the estimator. Columns (2) and (4) additionally include variables for payoff-irrelevant profile attributes. In the DirectBW/IndirectBW comparison, these controls indicate whether the Black profile rather than the White profile has tea as the stated drink preference, whether it shares the participant's stated preferences for sport, color, or drink, and whether it appears on the left side. In the IndirectBW/IndirectWW comparison, analogous controls are defined for the ambiguously framed option relative to the certain option. Task order is also included as a control. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Inattentive responding: excluding exceptionally fast participants

Participants were recruited via Prolific, which has been shown to be a suitable subject pool for social and economic science experiments and to deliver high data quality on

attention and related measures relative to common online alternatives (Palan and Schitter, 2018; Peer et al., 2022; Douglas et al., 2023; Gupta et al., 2021; Uittenhove et al., 2022). Nevertheless, I conduct a conservative duration-based robustness check by excluding participants with exceptionally fast completion times (bottom 5%; cutoff: 2.83 minutes). Appendix Figure B.2 shows the distribution of the survey duration and highlights the excluded left tail. Re-estimating the main specifications on the restricted sample yields qualitatively similar results, indicating that the main patterns are not driven by unusually fast and potentially low-engagement responses (see Table 4).

Table 4: Effects of race and ambiguous framing on allocation decisions: excluding the fastest 5% of participants

	Within-subject		Between-subject (first task only)	
Sample:	*DirectBW + IndirectBW*	*IndirectBW + IndirectWW*	*DirectBW + IndirectBW*	*IndirectBW + IndirectWW*
	(1)	(2)	(3)	(4)
IndirectBW	-0.070***	0.070***	-0.036*	0.107***
	(0.010)	(0.011)	(0.021)	(0.022)
Control mean	0.525	0.402	0.492	0.353
Participant fixed effects	Yes	Yes	–	–
Controls	Yes	Yes	Yes	Yes
Observations	8,208	8,208	2,032	2,008
Clusters (participants)	4,104	4,104	–	–

Notes: Each regression pools observations from two tasks only. This table re-estimates the main specifications after excluding the bottom 5% fastest participants based on total survey duration (participant-median duration cutoff: 170 seconds, 2.83 minutes). Columns (1)–(2) use within-subject variation (participant fixed effects; standard errors clustered by participant). Columns (3)–(4) use between-subject variation from each participant's first task only (heteroskedasticity-robust standard errors). Controls match the main specifications. In the within-subject columns, controls additionally include variables for payoff-irrelevant profile attributes. In the DirectBW/IndirectBW comparison, these controls indicate whether the Black profile rather than the White profile has tea as the stated drink preference, whether it shares the participant's stated preferences for sport, color, or drink, and whether it appears on the left side. In the IndirectBW/IndirectWW comparison, analogous controls are defined for the ambiguously framed option relative to the certain option. In the first-task columns, controls include age, gender, education, income, political ideology, and religion, as well as the same profile-composition and position controls. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Clarity of the payoff framing

A potential concern is that some participants may have interpreted the ambiguous framing ("the larger amount between 100 and 50 cents") as implying lower expected payoffs. To gauge comprehension of this wording, the survey included a post-task clarity question. As shown in Table 5, 88.4% of participants report that the framing was clear, 10.3% report that it was not clear, and 1.3% provided an open-ended text response. Within this

Table 5: Self-reported clarity of the ambiguous framing

Was the task wording clear?	Count	Share %
Clear	3,824	88.4%
Unclear	445	10.3%
Open-ended text response	58	1.3%
Total	4,327	1.000

Notes: The table reports the distribution of the question assessing whether the ambiguous framing ("the larger amount between 100 and 50 cents") was clear.

Table 6: Robustness to task wording: restricting to participants who report that the ambiguous framing was clear

	Within-subject		Between-subject (first task only)	
Sample:	*DirectBW + IndirectBW*	*IndirectBW + IndirectWW*	*DirectBW + IndirectBW*	*IndirectBW + IndirectWW*
	(1)	(2)	(3)	(4)
IndirectBW				
Full sample	-0.073***	0.070***	-0.039*	0.126***
	(0.010)	(0.010)	(0.020)	(0.021)
Clear wording only	-0.071***	0.071***	-0.042**	0.133***
	(0.010)	(0.011)	(0.021)	(0.023)
Control mean				
Full sample	0.526	0.401	0.496	0.350
Clear wording only	0.527	0.402	0.504	0.352
Participant fixed effects	Yes	Yes	–	–
Controls	Yes	Yes	Yes	Yes
Observations				
Full sample	8,654	8,654	2,145	2,117
Clear wording only	7,648	7,648	1,895	1,857
Clusters (participants)				
Full sample	4,327	4,327	–	–
Clear wording only	3,824	3,824	–	–

Notes: Each regression pools observations from two tasks only. "Clear wording only" restricts the sample to participants who report that the ambiguous framing ("the larger amount between 100 and 50 cents") was clear. Within-subject columns include participant fixed effects and standard errors clustered by participant. First-task columns use heteroskedasticity-robust standard errors. Controls match the corresponding main specifications. In the within-subject columns, controls additionally include variables for payoff-irrelevant profile attributes. In the DirectBW/IndirectBW comparison, these controls indicate whether the Black profile rather than the White profile has tea as the stated drink preference, whether it shares the participant's stated preferences for sport, color, or drink, and whether it appears on the left side. In the IndirectBW/IndirectWW comparison, analogous controls are defined for the ambiguously framed option relative to the certain option. In the first-task columns, controls include age, gender, education, income, political ideology, and religion, together with the same profile-composition and position controls. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

small subset of comments ($N = 56$), the most common themes included interpreting the wording as implying 100 cents, and describing the framing as "trick-like". Less frequent comments expressed uncertainty or mentioned 50-cent outcomes ($N = 10$). Although these interpretations were rare, they motivate the following robustness analysis, which examines whether the main results are sensitive to perceived framing ambiguity.

I re-estimate the main specifications restricting the sample to participants who report that the wording was clear. Table 6 shows that the estimates remain qualitatively unchanged, indicating that the main patterns are not driven by misunderstanding of the task wording.

Table 7: Effects of race and ambiguous framing on allocation decisions: robustness to profile drink preferences

	Within-subject		Between-subject (first task only)	
Sample:	*DirectBW + IndirectBW*	*IndirectBW + IndirectWW*	*DirectBW + IndirectBW*	*IndirectBW + IndirectWW*
	(1)	(2)	(3)	(4)
IndirectBW				
Full sample	-0.071***	0.055***	-0.034*	0.113***
	(0.010)	(0.010)	(0.021)	(0.021)
Same drink only	-0.113***	0.056***	-0.028	0.120***
	(0.024)	(0.019)	(0.032)	(0.031)
Participant fixed effects	Yes	Yes	–	–
Observations				
Full sample	8,654	8,654	2,145	2,117
Same drink only	1,528	2,128	871	1,014
Clusters (participants)				
Full sample	4,327	4,327	–	–
Same drink only	764	1,064	–	–

Notes: Each regression pools observations from two tasks only. "Same drink only" restricts the sample to choices in which the two compared options have the same stated drink preference. In the DirectBW/IndirectBW comparison, this means that the Black and White profiles both state their preferences for tea or for coffee. In the IndirectBW/IndirectWW comparison, it means that the ambiguously framed and certain options are either both tea or both coffee. Within-subject columns include participant fixed effects and standard errors clustered by participant. First-task columns use heteroskedasticity-robust standard errors. Full-sample rows report the corresponding unrestricted estimates for comparison. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Drink preference composition of profiles

Because the Stage 2 profiles correspond to real Stage 1 participants, payoff-irrelevant attributes such as tea versus coffee preference are not mechanically balanced across profiles. To assess whether the main results are driven by drink preferences rather than by ambiguous framing and race, I re-estimate the main comparisons on the subset of choices in which the two compared options have the same stated drink preference. In the DirectBW/IndirectBW comparison, this means that the Black and White profiles are

either both tea or both coffee; in the IndirectBW/IndirectWW comparison, it means that the ambiguously framed and certain options are either both tea or both coffee. In this restricted sample, the options compared have identical stated drink preferences. Hence, there is no within-choice variation in drink composition, and tea/coffee labels cannot explain the variation captured by the treatment effect.

Table 7 shows that the main results are qualitatively unchanged in the same-drink subsample. In the IndirectBW/IndirectWW comparison, the coefficient is 0.055 in the full sample and 0.056 in the same-drink subsample. In the DirectBW/IndirectBW comparison, the coefficient remains negative and statistically significant, changing from -0.071 in the full sample to -0.113 in the same-drink subsample. In the first-task analysis, the estimates remain directionally similar; the IndirectBW/IndirectWW comparison remains positive and statistically significant, while the DirectBW/IndirectBW estimate becomes less precise because the restriction substantially reduces the sample size. Overall, these results indicate that the main patterns are not driven by tea/coffee composition of the compared profiles.

Appendix D examines whether the attributes of the recipient profiles account for the results. Figure D.1 re-estimates both comparisons after exactly balancing recipient gender and both recipients' sport, color, and drink preferences across tasks. The DirectBW–IndirectBW effect remains negative, the IndirectBW–IndirectWW effect remains positive, and both remain statistically significant. Figure D.2 shows that omitting individual recipient profiles does not change the sign of either effect.

3.4 Heterogeneity of allocation choices

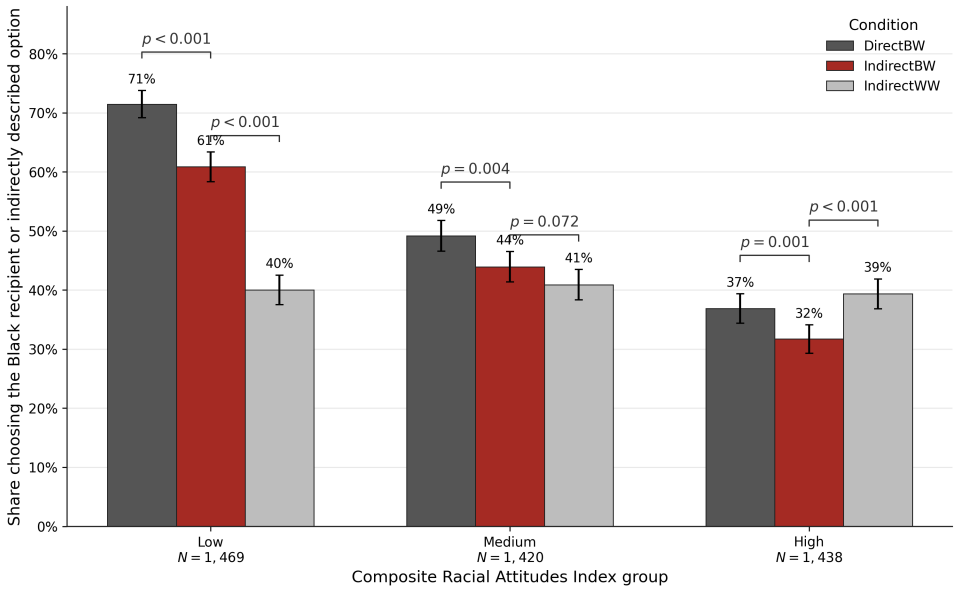
On average in the experimental sample, the results suggest that ambiguous framing does not broadly serve as moral cover for discrimination. However, the absence of a strong average treatment effect does not automatically suggest that discriminatory motives are absent. In fact, average effects may conceal heterogeneity. In this section, I explore whether patterns consistent with excuse-driven discrimination — using ambiguous framing as a cover to discriminate — exist for some subgroups; in particular, subgroups that hold more negative racial attitudes toward Black Americans.

I examine heterogeneity using the Composite Racial Attitudes Index[12]. Heterogeneity

[12]The Composite Racial Attitudes Index is constructed from the same six pre-registered items that also underlie the two component measures shown separately in the appendix: the two stereotype items form the Stereotype Index, and the four symbolic-racism items form the Symbolic Racism Index. Separate analyses using these two components yield the same qualitative pattern. Results for the Stereotype Index are presented in Appendix Figure G.3 and Table A.2, and the results for the Symbolic Racism Index are presented in Appendix Figure G.5 and Table A.3. Appendix Table G.1 shows the correlations among the Composite Racial Attitudes Index and its two underlying components, and Appendix Table G.2 shows

analyses based on the racial-attitude measures underlying the index were preregistered, although the preregistration did not specify combining these measures into a single composite index. Figure 2 shows substantial baseline heterogeneity. In the Low composite-attitudes group, participants choose the Black recipient in about 71 percent of baseline BW choices (DirectBW), whereas in the High composite-attitudes group this falls to about 37 percent. Thus, baseline avoidance of the Black recipient is concentrated among respondents who report holding more negative racial attitudes. Table 8, Panel A, shows that attaching ambiguous framing to the Black-recipient option (IndirectBW) lowers Black choice on average, but this reduction becomes smaller as racial attitudes become more negative. Although the absolute percentage-point reductions vary across the Composite Racial Attitudes Index groups, the reductions relative to each group's DirectBW baseline are similar: 14.9% in the Low group, 10.6% in the Medium group, and 14.0% in the High group. This suggests that the smaller absolute reduction among participants with more negative racial attitudes largely reflects their lower starting probability of choosing the Black recipient, rather than an unusually small proportional response to ambiguous framing.

Figure 2: Mean allocation choices by Composite Racial Attitudes Index groups: Low, Medium, High



Notes: The y-axis pools two distinct outcomes: for DirectBW and IndirectBW bars, it shows the share of participants choosing the Black-recipient option; for IndirectBW and IndirectWW bars, it shows the share choosing the ambiguously framed option. Error bars are 95% confidence intervals around the mean. N denotes unique participants in each Composite Racial Attitudes Index group. The Composite Racial Attitudes Index is an item-weighted average of six racial-attitude items, with higher values indicating more negative attitudes toward Black Americans. The p-values are from paired tests within each bin.

To examine more directly whether allocation choices depend on the race of the recipient

pairwise correlations among the six items used to construct the composite measure.

Table 8: Heterogeneity of allocation choices by the Composite Racial Attitudes Index

Sample:	Panel A *DirectBW + IndirectBW*	Panel B *IndirectBW + IndirectWW*
	(1)	(2)
IndirectBW	-0.1154***	0.2914***
	(0.0181)	(0.0188)
IndirectBW × Composite Racial Attitudes Index	0.1133***	-0.5884***
	(0.0404)	(0.0415)
Participant fixed effects	Yes	Yes
Choice-level controls	Yes	Yes
Participants	4,327	4,327
Observations	8,654	8,654

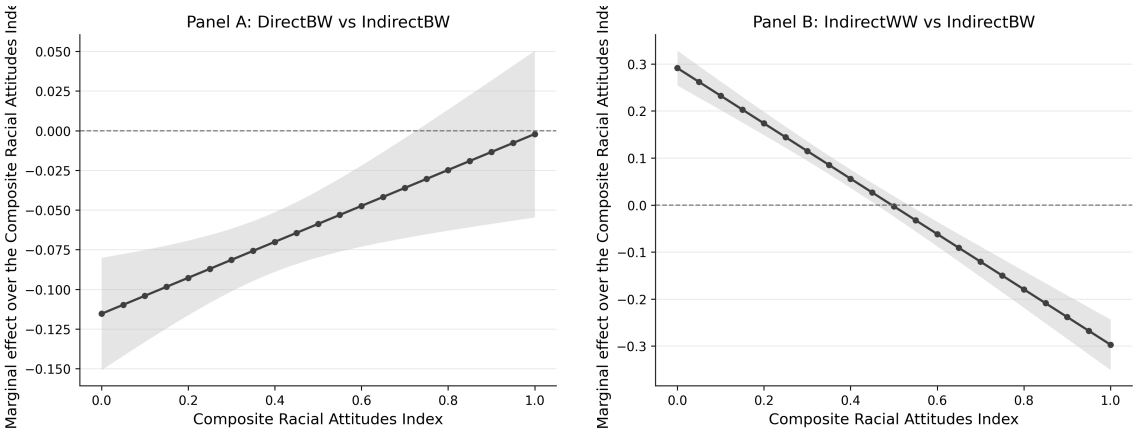
Notes: Each regression pools observations from two tasks only. The dependent variable equals 1 if the Black-recipient option is selected in the DirectBW/IndirectBW comparison (Panel A), and 1 if the ambiguously framed option is selected in the IndirectBW/IndirectWW comparison (Panel B). Specifications use participant fixed effects with participant-clustered standard errors and the same choice-level controls as in the main specification. The Composite Racial Attitudes Index is the raw 0–1 index, with higher values indicating more negative attitudes toward Black Americans. Because the index is not centered, the IndirectBW coefficient is the estimated effect when the index equals zero. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

who benefits from ambiguous framing, I compare IndirectBW with IndirectWW. The payoff descriptions are held fixed, while the ambiguous option benefits a Black recipient in IndirectBW and a White recipient in IndirectWW. Table 8, Panel B, shows a negative and statistically significant interaction between IndirectBW and the Composite Racial Attitudes Index. This implies that as racial attitudes become more negative, participants become less willing to choose the ambiguous option when it benefits a Black recipient rather than a White recipient. Figure 2 illustrates this relationship. In the Low composite-attitudes group, approximately 61% choose the ambiguous option when it benefits the Black recipient, compared with approximately 40% when it benefits the White recipient. This pattern reverses in the High group: 39.4% choose the ambiguous option in IndirectWW, compared with 31.7% in IndirectBW, a difference of 7.6 percentage points ($p < 0.001$). Figure 3 shows how this difference changes over the raw Composite Racial Attitudes Index.

Taken together, these results show that the relationship between ambiguous framing and allocation choices varies with both racial attitudes and the race of the recipient who benefits from the ambiguous option. At the level of group means, the pattern in the High composite-attitudes group moves in the direction predicted by the moral-cover explanation.

Exploratory analysis of participants' joint choices, presented in Appendix C, weakens this interpretation. The High-group difference is driven by participants who choose the White recipient in DirectBW, rather than by those who choose the Black recipient in DirectBW but the White recipient in IndirectBW. Among participants who choose the

Figure 3: Marginal effects over the Composite Racial Attitudes Index



Notes: Solid lines plot marginal effects from participant fixed-effects regressions with a linear interaction between treatment and the raw Composite Racial Attitudes Index. Shaded bands are 95% confidence intervals with standard errors clustered at the participant level. Higher values of the index indicate more negative attitudes toward Black Americans. All specifications include the same choice-level controls as in the main specification. The dashed horizontal line marks zero.

Black recipient in DirectBW, more negative racial attitudes predict choosing the White recipient in IndirectBW, but this relationship is similar whether they accept or reject the indirect description in IndirectWW. Thus, allocation choices vary with racial attitudes and with the race of the recipient who benefits from the ambiguous wording, but the evidence does not establish that this wording reveals discrimination hidden in the baseline task.

Because racial attitudes were elicited after the allocation tasks, the responses could partly reflect consistency with earlier choices, rationalization of the choices, or experimental demand effects. As a complementary exploratory analysis, I examine heterogeneity by political views, which were elicited before the tasks. This analysis is motivated by evidence linking political conservatism to stronger implicit and explicit preferences for White over Black Americans (Nosek et al., 2007, 2009). Related research shows that beliefs about racial discrimination differ between Republicans and Democrats (Haaland and Roth, 2023), and that responses to Black and White tutors differ by political orientation (Bartoš et al., 2025). Appendix Figure B.3 shows a similar pattern to the racial-attitude results[13]: liberal participants choose the ambiguously framed option more often when it benefits the Black recipient than when it benefits the White recipient, whereas conservative participants show the reverse.

[13]The mean Composite Racial Attitudes Index is 0.24 among liberals, 0.40 among moderates, and 0.49 among conservatives. Differences between these three groups account for 24.8% of its variation ($\eta^2 = 0.248$; $N = 4{,}277$, excluding 50 apolitical respondents).

4 Conclusion

This paper asks whether ambiguous framing in a decision environment can provide cover for racial discrimination by allowing a discriminatory choice to appear to result from a plausible error. To study this question, I run a pre-registered incentivized experiment with 4,327 White U.S. participants in which monetary payoffs are held fixed and the framing of the payoff description and the race associated with the ambiguous option vary across conditions. This design allows me to distinguish between general tendency to avoid ambiguously framed options and the selective exploitation of ambiguous framing as a justification device against Black recipients.

The main finding is that ambiguous framing does not, on average, enable excuse-driven discrimination. The more informative comparison — between two equally ambiguous tasks that attach the ambiguous option to a Black or a White recipient — directly contradicts the moral cover hypothesis. Participants choose the ambiguously framed option more often when it benefits the Black recipient — the opposite of what selective exploitation of ambiguous framing would predict — pointing to general tendency to avoid ambiguous wording rather than excuse-driven discrimination.

At the same time, the aggregate result masks heterogeneity. Participants with more negative racial attitudes are less likely to choose the ambiguous option when it benefits the Black recipient than when it benefits the White recipient. However, the exploratory analysis of individual choice patterns weakens the moral-cover interpretation of this result. The difference is driven by participants who choose the White recipient when the choice is transparent, rather than by participants who choose the Black recipient in DirectBW but the White recipient when ambiguous framing is attached to the Black-recipient option. Among participants who do choose the Black recipient in DirectBW, more negative racial attitudes predict choosing the White recipient in IndirectBW. But this relationship appears both among participants who accept ambiguous framing when it benefits a White recipient in IndirectWW and among those who reject it. The result therefore does not distinguish excuse-driven discrimination from a more general tendency to avoid ambiguous options.

One possible explanation is that the experimental setting creates relatively little reputational pressure or other costs to conceal discrimination. Excuse-driven discrimination requires some cost of expressing a discriminatory preference openly; that cost is what makes a justification valuable. In this study, choices are anonymous and participants are not required to explain them to recipients or other participants. This may reduce the need for a justification, although it does not eliminate possible internal moral or self-image costs. Consistent with this, participants in the High composite-attitudes group already

choose the White recipient in 63.1% of DirectBW choices, without any ambiguous framing.

Two implications follow. First, ambiguous framing does not on its own enable excuse-driven discrimination. A general tendency to avoid ambiguously framed options dominates in the average results, and the subgroup reversal does not come from the individual choice pattern predicted by the moral-cover explanation. Second, group-level heterogeneity can resemble moral cover even when individual choice patterns do not clearly follow the mechanism. Subgroup means alone are therefore weak evidence for excuse-driven discrimination. Ambiguous framing remains common in subjective performance evaluations, discretionary hiring decisions, and credit and housing markets. Future work could test the mechanism more directly by independently varying the framing and whether decisions are observed, must be explained, or carry consequences for the decision-maker.

Appendix A: Tables

Table A.1: Descriptive statistics (N = 4,327)

Numeric (mean ± SD)	
Age (years)	40.25 ± 11.29
Survey duration	7 min 20 sec ± 6 min 1 sec
Age group	
18–24	310 (7.2%)
25–34	1241 (28.7%)
35–44	1212 (28.0%)
45–54	979 (22.6%)
55–64	585 (13.5%)
Gender	
Female	2329 (53.8%)
Male	1957 (45.2%)
Other	41 (0.9%)
Political ideology	
Very conservative	584 (13.5%)
Conservative	1139 (26.3%)
Moderate	900 (20.8%)
Liberal	1057 (24.4%)
Very liberal	597 (13.8%)
Apolitical	50 (1.2%)
Highest education	
< HS	27 (0.6%)
HS diploma	435 (10.1%)
Some college	713 (16.5%)
Associate	328 (7.6%)
Bachelor's	1668 (38.5%)
Master's	947 (21.9%)
Doctorate	209 (4.8%)
Household income	
< $10 k	145 (3.4%)
$10-14 k	120 (2.8%)
$15-24 k	222 (5.1%)
$25-34 k	308 (7.1%)
$35-49 k	468 (10.8%)
$50-74 k	808 (18.7%)
$75-99 k	715 (16.5%)
$100-149 k	928 (21.4%)

$150-199 k	389 (9.0%)
≥ $200 k	224 (5.2%)
Marital status	
Married	2304 (53.2%)
Single, never married	1445 (33.4%)
Divorced	312 (7.2%)
Civil union	139 (3.2%)
Widowed	69 (1.6%)
Separated	58 (1.3%)
Number of children	
0	1770 (40.9%)
1	742 (17.1%)
2	1114 (25.7%)
3	472 (10.9%)
4 or more	229 (5.3%)
Occupation	
FT employed	2481 (57.3%)
PT employed	718 (16.6%)
Self-employed	388 (9.0%)
Unemployed	328 (7.6%)
Homemaker	201 (4.6%)
Retired	115 (2.7%)
Student	96 (2.2%)
Sexual orientation	
Heterosexual	3559 (82.3%)
Bisexual	480 (11.1%)
Homosexual	200 (4.6%)
Other	88 (2.0%)
Religion	
Christian	2677 (61.9%)
Not religious	1380 (31.9%)
Jewish	61 (1.4%)
Muslim	34 (0.8%)
Buddhist	17 (0.4%)
Hindu	9 (0.2%)
Other	149 (3.4%)

Table A.2: Heterogeneity of allocation choices by the Stereotype Index

Sample:	Panel A *DirectBW + IndirectBW*	Panel B *IndirectBW + IndirectWW*
	(1)	(2)
IndirectBW	-0.1015***	0.1740***
	(0.0200)	(0.0202)
IndirectBW × Stereotype Index	0.0115*	-0.0441***
	(0.0063)	(0.0065)
Participant fixed effects	Yes	Yes
Participants	4,327	4,327
Observations	8,654	8,654

Notes: Each regression pools observations from two tasks only. The dependent variable equals 1 if the Black-recipient option is selected in the DirectBW/IndirectBW comparison (Panel A), and 1 if the ambiguously framed option is selected in the IndirectBW/IndirectWW comparison (Panel B), where the ambiguous option is attached to the Black recipient in IndirectBW and to the White recipient in IndirectWW. Specifications use participant fixed effects with participant-clustered standard errors. The Stereotype Index is the raw index. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

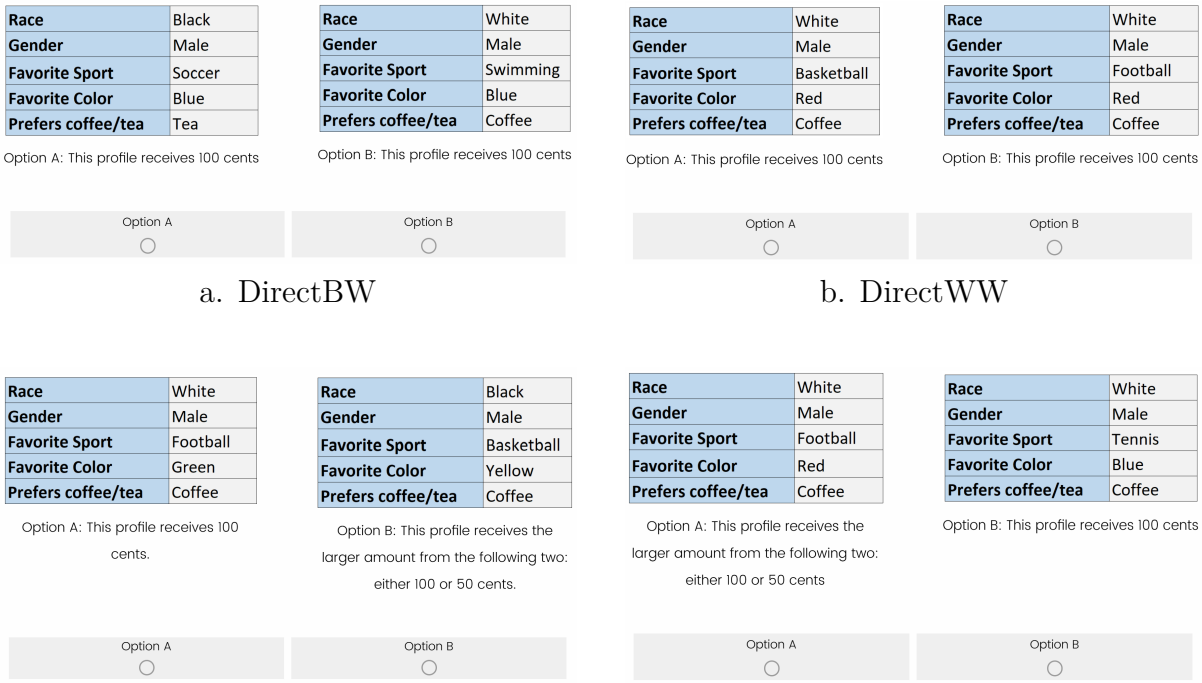
Table A.3: Treatment effect heterogeneity by the Symbolic Racism Index

Sample:	DirectBW + IndirectBW *DirectBW + IndirectBW*		IndirectBW + IndirectWW *IndirectBW + IndirectWW*	
	(1)	(2)	(3)	(4)
IndirectBW	-0.070***	-0.073***	0.055***	0.071***
	(0.010)	(0.010)	(0.010)	(0.010)
IndirectBW × Symbolic Racism Index	0.021**	0.024***	-0.129***	-0.128***
	(0.009)	(0.009)	(0.009)	(0.009)
Controls	No	Yes	No	Yes
Participant fixed effects	Yes	Yes	Yes	Yes
Observations	8,654	8,654	8,654	8,654
Clusters (participants)	4,327	4,327	4,327	4,327

Notes: Each regression pools observations from two tasks only. The estimation uses linear probability models with participant fixed effects and standard errors clustered at the participant level. The Symbolic Racism Index is standardized to mean zero and unit standard deviation. Columns (1)–(2) use the DirectBW and IndirectBW sample; the dependent variable equals 1 if the Black profile is chosen. Columns (3)–(4) use the IndirectBW and IndirectWW sample; the dependent variable equals 1 if the ambiguously framed option is chosen. Controls in columns (2) and (4) additionally include task-order controls and variables for payoff-irrelevant profile attributes. In the DirectBW/IndirectBW comparison, these controls indicate whether the Black profile rather than the White profile has tea as the stated drink preference, whether it shares the participant's stated preferences for sport, color, or drink, and whether it appears on the left side. In the IndirectBW/IndirectWW comparison, analogous controls are defined for the ambiguously framed option relative to the certain option. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix B: Figures

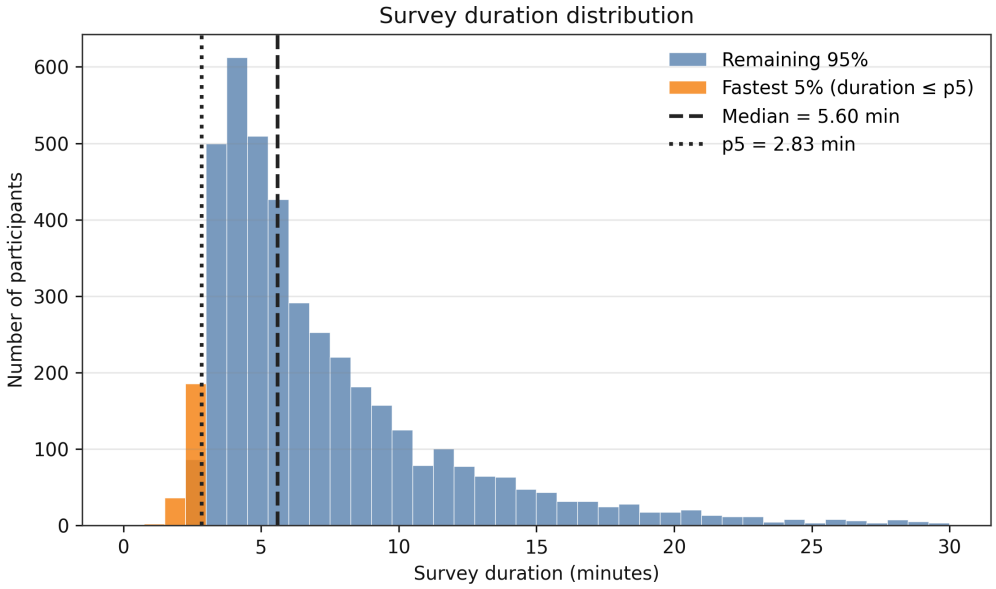
Figure B.1: Task screenshots



a. DirectBW b. DirectWW

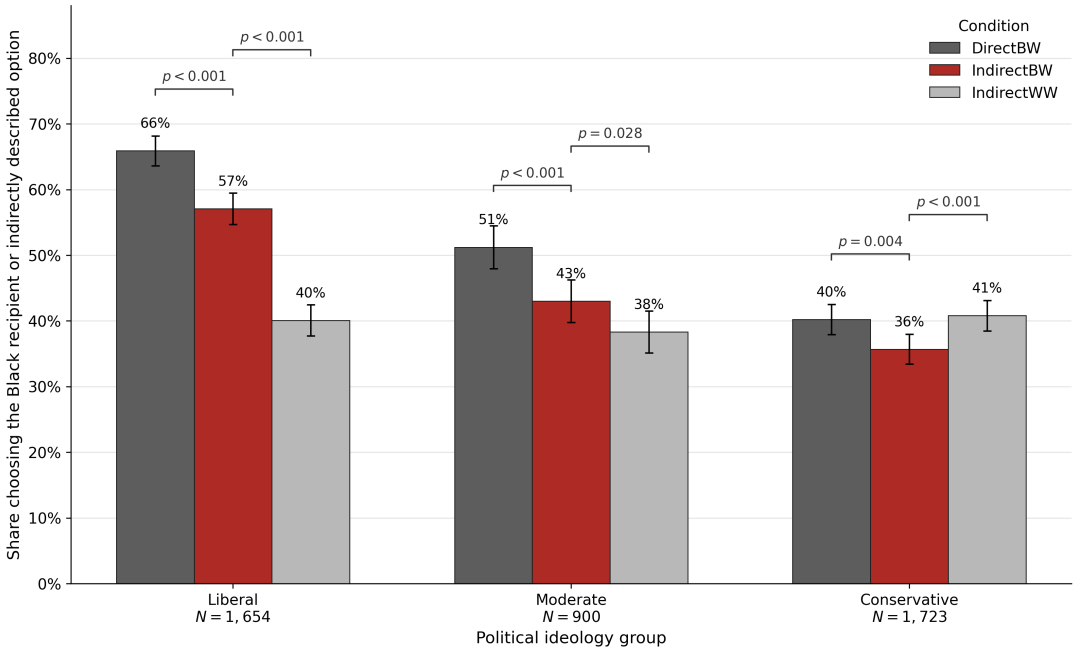
c. IndirectBW d. IndirectWW

Figure B.2: Survey duration distribution



Notes: The figure plots the distribution of participant-level survey duration in minutes. Orange bars highlight the bottom 5% fastest participants (duration $\leq$ p5), which defines the exclusion used in the robustness analysis. The dashed vertical line indicates the sample median (5.60 minutes) and the dotted line indicates the p5 cutoff (2.83 minutes). For readability, the histogram is truncated at 30 minutes; 24 participants (0.55%) took longer than 30 minutes.

Figure B.3: Mean allocation choices by political views



Notes: DirectBW and IndirectBW bars show the share choosing the Black-recipient option; IndirectWW bars show the share choosing the ambiguously framed White-recipient option. In IndirectBW, choosing the Black recipient also means choosing the ambiguously framed option. Error bars show 95% confidence intervals around the means. Bracketed p-values are from paired tests within each political group. Liberal combines liberal and very liberal participants; Conservative combines conservative and very conservative participants. The 50 apolitical participants are excluded. N denotes participants in each group. Political views were elicited before the allocation tasks. This analysis is exploratory.

Appendix C: Exploratory analysis of individual choice patterns

The group means in Section 3.4 do not show whether the 7.6 percentage-point difference in the High composite-attitudes group is driven by participants whose choices across all three tasks follow the pattern consistent with excuse-driven discrimination. I examine this possibility using participants' joint choice patterns. This analysis was not specified in the preregistration and should therefore be interpreted as exploratory.

Does the pattern in the High composite-attitudes group imply excuse-driven discrimination?

If the 7.6 percentage-point difference reflects excuse-driven discrimination, it should be concentrated among participants who choose the Black-recipient option in DirectBW but the White-recipient option in IndirectBW. Choosing the Black recipient in the transparent task means that any discriminatory preference stays latent there. Choosing the White recipient when the Black-recipient option is described ambiguously is what the excuse-driven account predicts: the wording supplies the justification. To contribute to the 7.6 percentage-point difference, these participants must also choose the ambiguous option in IndirectWW, where it benefits a White recipient.

The data do not show this pattern. Among the 530 High-group participants who choose the Black recipient in DirectBW, 19.1% choose the ambiguously framed option only when it benefits the White recipient and 20.0% choose it only when it benefits the Black recipient. The difference is -0.9 percentage points ($p = 0.781$). Among the 908 who choose the White recipient in DirectBW, the same difference is 12.7 percentage points ($p < 0.001$). The aggregate 7.6 percentage-point difference is therefore driven by participants who already favor the White recipient in the transparent task.

Among participants who choose the Black recipient in DirectBW, does the choice of the White recipient in IndirectBW vary with racial attitudes?

The pattern consistent with latent bias starts with choosing the Black recipient in DirectBW. I therefore take all 2,278 participants who make that choice and ask whether more negative racial attitudes predict choosing the White-recipient option in IndirectBW. Figure C.1 and Table C.1 show that a one-standard-deviation increase in the Composite Racial Attitudes Index is associated with a 10.9 percentage-point increase in that probability ($p < 0.001$).

I then split these participants by their choice in IndirectWW, because that choice changes

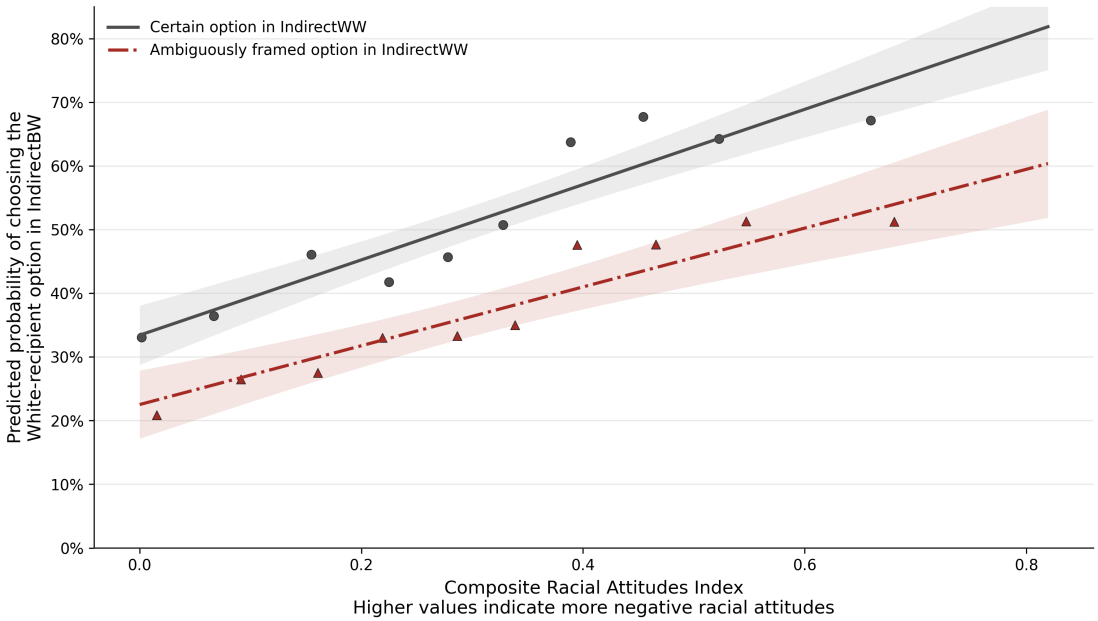
what choosing the White recipient in IndirectBW means. Participants who choose the ambiguous option in IndirectWW and the White-recipient option in IndirectBW accept ambiguously framed option when it benefits a White recipient and reject it when it benefits a Black recipient. This is the pattern most closely aligned with excuse-driven discrimination. For them, a one-standard-deviation increase in the index is associated with a 9.5 percentage-point increase in the probability of choosing the White-recipient option in IndirectBW (Holm-adjusted $p < 0.001$). Participants who choose the certain option in IndirectWW and the certain White-recipient option in IndirectBW take the certain option in both tasks. Here the two motives are confounded: avoiding ambiguous framing and favoring the White recipient point to the same choice. For them, the estimate is 12.1 percentage points (Holm-adjusted $p < 0.001$), and it is not statistically distinguishable from 9.5 ($p = 0.193$). Because both subgroups are defined by realized choices rather than by random assignment and start from different base rates of choosing the White recipient in IndirectBW, I read this comparison as descriptive rather than causal. More negative racial attitudes predict choosing the White-recipient option in IndirectBW regardless of the choice in IndirectWW, and the relationship is no stronger in the subgroup whose behavior fits excuse-driven discrimination. Figure C.2 illustrates this point: both the pattern consistent with moral cover and choosing the certain option in both tasks become more common as racial attitudes become more negative.

Table C.1: Racial attitudes and choice of the White-recipient option in IndirectBW

	All participants who chose Black in DirectBW	Choice in IndirectWW		
Specification	Pooled	Certain White-recipient option	Ambiguous White-recipient option	Difference between the two slopes
Baseline	0.109***	0.121***	0.095***	−0.027
	(0.010)	(0.013)	(0.016)	(0.020)
Full controls	0.079***	0.091***	0.063***	−0.029
	(0.013)	(0.015)	(0.018)	(0.021)
Task-order indicators	Yes	Yes	Yes	Yes
Participants, baseline specifications	2,278	2,278	2,278	2,278
Participants, full-control specification	–	2,243	2,243	2,243

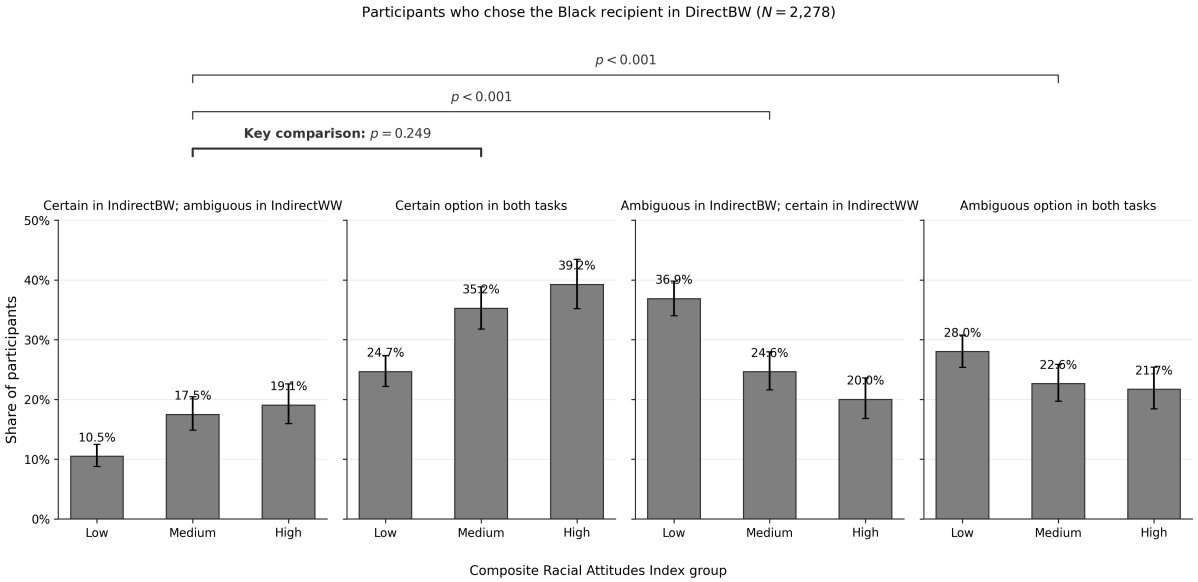
Notes: The sample is restricted to participants who chose the Black recipient in DirectBW. The dependent variable equals one when a participant chooses the certain White-recipient option rather than the ambiguous Black-recipient option in IndirectBW. Entries are coefficients from linear probability models; HC3 robust standard errors are in parentheses. The Composite Racial Attitudes Index is standardized within this sample, so each coefficient is the change in probability associated with a one-standard-deviation increase in negative self-reported racial attitudes. The pooled estimates come from models that do not condition on the IndirectWW choice. The other columns come from models that interact the racial-attitudes index with the IndirectWW choice. The final column is the slope for participants choosing ambiguous framing in IndirectWW minus the slope for participants choosing the certain option. For the three baseline conditional tests, Holm-adjusted p-values are < 0.001, < 0.001, and 0.193, respectively. The full-control specification includes the paper's demographic, recipient-gender, profile-composition, screen-position, and task-order controls. The difference between the two full-control slopes has $p = 0.168$. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.1$.

Figure C.1: Choice of the White-recipient option in IndirectBW over the Composite Racial Attitudes Index



Notes: The sample is restricted to participants who choose the Black recipient in DirectBW ($N = 2{,}278$). The outcome equals one when the participant chooses the certain White-recipient option rather than the ambiguous Black-recipient option in IndirectBW. Lines show average predicted probabilities from a linear probability model with the standardized Composite Racial Attitudes Index, the IndirectWW choice, their interaction, and indicators for all 24 randomized task sequences. Predictions average over the observed task-order distribution. Shaded areas are 95% confidence intervals based on HC3 robust standard errors. Points show raw shares within approximately equal-sized index groups, calculated separately by the IndirectWW choice. The horizontal axis displays the raw 0–1 index over its first to ninety-ninth percentile range. Higher values indicate more negative self-reported racial attitudes.

Figure C.2: Joint choice patterns in IndirectBW and IndirectWW



Notes: Notes: The sample comprises participants who chose the Black recipient in DirectBW (N = 2,278). Bars show observed percentages within the Low, Medium and High Composite Racial Attitudes Index groups; higher values indicate more negative self-reported racial attitudes. Within each group, the four patterns sum to 100% before rounding. Whiskers show 95% Wilson confidence intervals. Brackets compare the first pattern with each of the other three patterns. The p-values test whether the relative probability of the two connected patterns changes with the continuous index, rather than whether their shares are equal within an attitude group. Tests are two-sided Wald tests from a multinomial logit model with the index standardized within the selected sample, task-sequence indicators and heteroskedasticity-robust standard errors. Holm adjustment covers all six pairwise pattern comparisons, of which three are displayed. The highlighted bracket compares the first pattern with choosing the certain option in both tasks.

Appendix D: Profile pool composition and exposure across conditions

Recipient pool.

The Stage 2 recipient profiles were constructed from real participants recruited in Stage 1, so allocation decisions have consequences for actual people. Of the 50 Stage 1 participants, two identified as Hispanic or Latino and were not included in the Black–White recipient pool. The final pool contains 48 profiles: 16 Black and 32 White. Because the profiles were drawn from actual Stage 1 participants, payoff-irrelevant profile attributes are not mechanically balanced by race in the underlying profile pool. Some non-racial attributes shown to participants may therefore be correlated with race. Table D.1 describes the composition of the profiles across conditions. Table D.2 shows the distribution of sport, color, and drink preferences among Black and White profiles in the recipient pool.

Adjustment in the main regressions.

In the main regressions, I address these differences by controlling for recipient attributes, expressed as differences between the two recipients: their tea/coffee preferences and whether each recipient shares the participant's sport, color, and drink preferences. The regressions also control for the position of the option.

Alternative adjustments.

Figure D.1 compares the main estimates with two alternatives. The first reweights choices so that the compared tasks have the same joint distribution of both recipients' sport, color, and drink preferences within each recipient-gender arm, and the same weighted shares of the male and female arms. The second applies double machine learning using the same balanced profile composition. Logistic regression or gradient-boosted trees predict (i) which task a choice comes from and (ii) the probability of choosing the relevant option in each task. Specifications with additional controls also include participant characteristics and preferences, task order, and whether the relevant option appears on the left or right.

The balanced samples.

Exact balancing retains only attribute combinations that occur in both compared tasks, so coverage differs between the two comparisons. For DirectBW vs. IndirectBW (Figure D.1 Panel A), all 190 combinations occur in both tasks, and balancing retains all 8,654 choices from 4,327 participants. For IndirectBW vs. IndirectWW (Figure D.1 Panel B), only 61 of 471 combinations occur in both tasks. Balancing retains 2,464 choices

from 2,087 participants, of whom only 377 contribute choices from both tasks. The balanced estimate in Panel B therefore covers fewer attribute combinations and relies less on within-participant comparisons.

Omitting individual profiles.

Figure D.2 tests whether the results depend on individual recipients. Each of the 48 profiles is omitted in turn, participants who encountered it in either relevant task are excluded, and both comparisons are re-estimated. Both overall estimates keep their direction in all 48 omissions. The indirect estimate remains significant at the 5% level in all 48 paired specifications and in 44 of the 48 exact-balancing specifications.

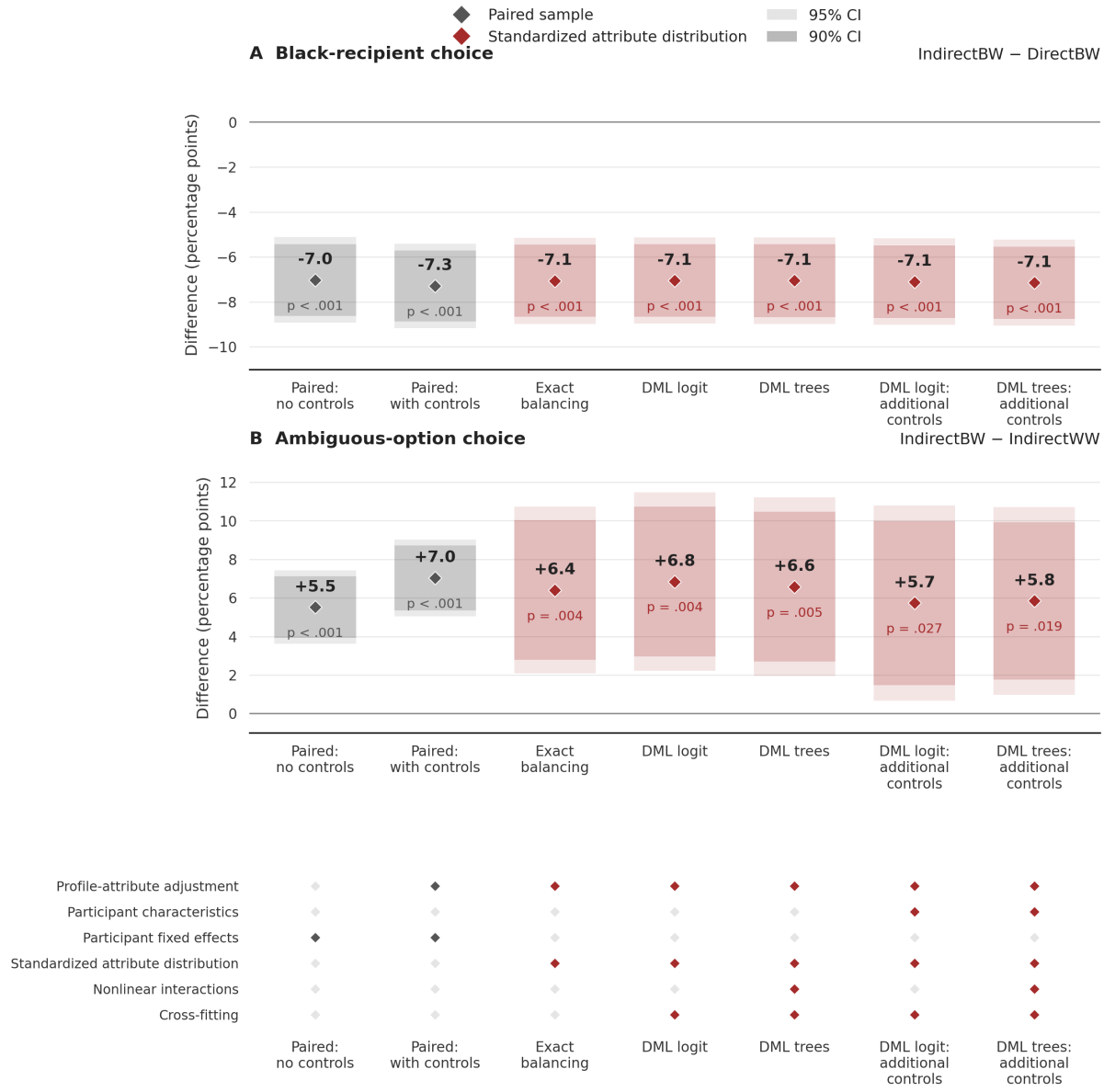
These checks indicate that the results are robust to observed profile composition within the experimental recipient pool. Statistical inference is conditional on this pool, and generalizing to a broader population of recipient profiles would require additional assumptions.

Table D.1: Composition of profiles displayed across conditions

Condition	Displayed profile	Tea share	Sport mode (share)	Color mode (share)	Δ Tea share	Sport dist. p	Color dist. p
DirectBW	Black profile	0.690	Basketball (0.380)	Blue (0.441)			
DirectBW	White profile	0.282	Football (0.350)	Blue (0.247)			
DirectBW	Difference (Black–White)				0.407	< 0.001	< 0.001
IndirectBW	Black profile	0.680	Basketball (0.384)	Blue (0.429)			
IndirectBW	White profile	0.284	Football (0.358)	Blue (0.249)			
IndirectBW	Difference (Black–White)				0.396	< 0.001	< 0.001
IndirectWW	White profile (Ambiguous)	0.299	Football (0.358)	Blue (0.246)			
IndirectWW	White profile (Non-ambiguous)	0.285	Football (0.352)	Blue (0.260)			
IndirectWW	Difference (Ambiguous–Non-ambiguous)				0.014	0.977	0.511

Notes: Each condition has $N = 4{,}327$ observations. This table summarizes the composition of the profiles shown to participants in each condition. Because the profiles are based on real individuals recruited in Stage 1, non-racial attributes (tea/coffee, sport, color) are fixed and may be correlated with race in the underlying profile pool. "Tea share" is the fraction of displayed profiles with a stated preference for tea rather than coffee. "Sport mode" and "Color mode" report the most common category and its share in parentheses. "Δ Tea share" reports the difference in tea share between the two profiles compared within a condition. "Sport dist. p" and "Color dist. p" report χ^2 test p-values for equality of the full categorical distributions across the two displayed profiles within a condition (Black vs. White in DirectBW and IndirectBW; ambiguous vs. non-ambiguous White profiles in IndirectWW). The main regressions account for the recipients' tea/coffee preferences and for whether each recipient shares the participant's sport, color, and drink preferences, expressed as differences between the two recipients. They also control for option position.

Figure D.1: Overall results after adjustment for recipient attributes



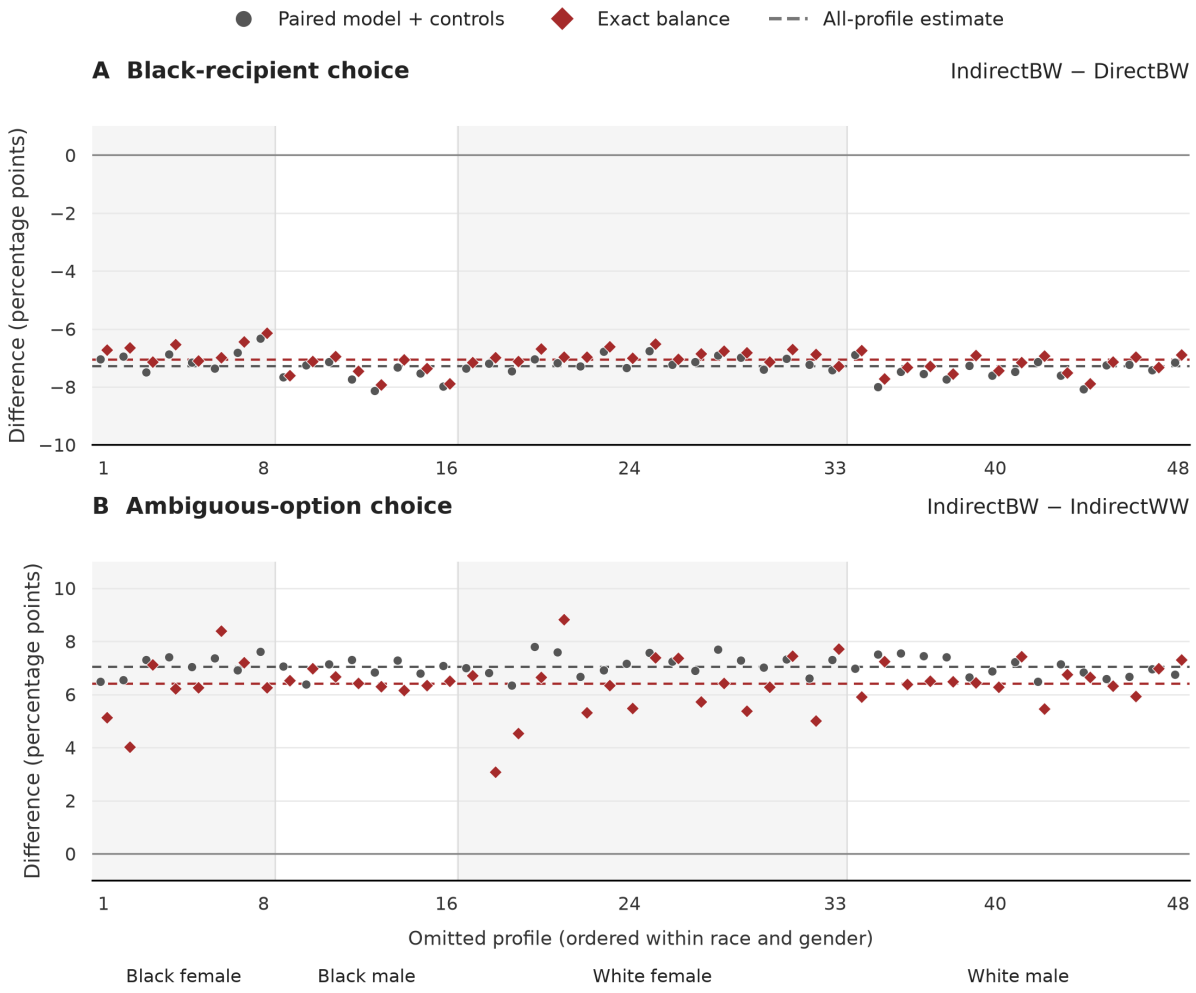
Notes: Panel A reports IndirectBW minus DirectBW in Black-recipient choice; Panel B reports IndirectBW minus IndirectWW in ambiguous-option choice. Estimates are in percentage points. Diamonds denote coefficients; light and dark bands show 95% and 90% confidence intervals, respectively. Both paired specifications include participant fixed effects; *with controls* adds the paper's profile-composition and option-position controls. *Exact balancing* equalizes the joint distribution of recipient gender and both recipients' sport, color, and drink preferences across tasks using empirical matching weights (Li and Greene, 2013). *DML logit* and *DML trees* use logistic regression and gradient-boosted trees, respectively, implemented in DoubleML (Chernozhukov et al., 2018; Bach et al., 2022). These models include displayed recipient attributes; *additional controls* adds participant characteristics and preferences, task order, and option position. Cross-fitting uses five participant-level folds and three repetitions. Exact balancing and DML use the same standardized attribute distribution, retaining 8,654 direct-comparison choices and 2,464 indirect-comparison choices from 2,087 participants. Filled matrix diamonds indicate included features. Standard errors account for repeated choices by participants; p-values test a zero contrast. Balanced and full-sample paired estimates may represent different profile compositions. Inference is conditional on the experimental recipient pool and the empirical support and balancing weights.

Table D.2: Distribution of payoff-irrelevant profile attributes by race in the profile pool

Category	Black profiles N (%)	White profiles N (%)
Panel A. Favorite sport		
Football	3 (19)	11 (34)
Basketball	6 (38)	6 (19)
Swimming	2 (12)	6 (19)
Soccer	3 (19)	2 (6)
Baseball	1 (6)	3 (9)
Tennis	1 (6)	3 (9)
Running	0 (0)	1 (3)
Panel B. Favorite color		
Blue	7 (44)	8 (25)
Green	1 (6)	6 (19)
Pink	3 (19)	4 (12)
Black	3 (19)	3 (9)
Purple	1 (6)	5 (16)
Red	0 (0)	4 (12)
Yellow	1 (6)	1 (3)
Gray	0 (0)	1 (3)
Panel C. Tea/coffee preference		
Coffee	5 (31)	23 (72)
Tea	11 (69)	9 (28)

Notes: The profile pool contains $N_{\text{Black}} = 16$ Black profiles and $N_{\text{White}} = 32$ White profiles. Because these profiles correspond to real Stage 1 participants rather than experimentally constructed profiles, payoff-irrelevant attributes are not mechanically balanced by race and may therefore be correlated with it. The main regressions account for the recipients' tea/coffee preferences and for whether each recipient shares the participant's sport, color, and drink preferences, expressed as differences between the two recipients. They also control for option position.

Figure D.2: Sensitivity to individual recipient profiles



Notes: Each point reports the estimated contrast after omitting one of the 48 recipient profiles. Participants are excluded if the omitted profile appears in either relevant task, preserving complete pairs in the paired specification. Panel A reports IndirectBW minus DirectBW in Black-recipient choice; Panel B reports IndirectBW minus IndirectWW in ambiguous-option choice. Estimates are in percentage points. Circles denote the paired specification with profile-composition and position controls; diamonds denote exact balancing. Dashed lines show the corresponding estimates using all profiles. Common support and balancing weights are recomputed after each omission, so the matched target may change. Confidence intervals are omitted. In Panel B, all estimates remain positive; the contrast is significant at 5% in 48 of 48 paired checks and 44 of 48 exact-balancing checks. DML is not re-estimated. Inference is conditional on the experimental recipient pool.

Appendix E: Supplementary analyses using the DirectWW task

I use the DirectWW task as a race-neutral placebo task for two supplementary purposes. First, I use it to assess whether generic left-right choice patterns differ between the two simple tasks, DirectWW and DirectBW. Second, I use it as a neutral baseline for analyzing task duration.

Left-right choice patterns

The first diagnostic is whether participants are more likely to choose the left option in the simple Black–White task than in the neutral White–White placebo task. The share choosing the left option is 52.0% in DirectWW and 50.4% in DirectBW. The participant-level paired difference, defined as $DirectBW - DirectWW$, equals -0.016 (SE $= 0.011$), indicating that the left-choice rate is 1.6 percentage points lower in DirectBW, but the difference is not statistically significant. Thus, there is no strong evidence that the simple Black–White framing meaningfully shifts generic left-right choice probability relative to the neutral placebo task.

Task duration

I next use DirectWW as a neutral baseline for analyzing task duration. As shown in Table E.1, average task duration is 15.1 seconds in DirectWW, 15.7 seconds in DirectBW, 19.1 seconds in IndirectBW, and 19.8 seconds in IndirectWW. Relative to DirectWW, the difference for DirectBW is small, about 0.6 seconds, and not statistically significant ($p = 0.261$). By contrast, both ambiguously framed tasks take substantially longer than DirectWW: IndirectBW by about 4.0 seconds and IndirectWW by about 4.8 seconds (both $p < 0.001$). The difference between the two ambiguous tasks is about 0.7 seconds and is not statistically significant ($p = 0.203$). Overall, these results also point to a general tendency to avoid ambiguously framed options, rather than a uniquely stronger time cost when ambiguous framing is attached to the Black recipient.

Table E.1: Task duration (seconds) by conditions

Condition	Mean	Median	SD	Diff vs. DirectWW	p-value
DirectWW	15.06	10.04	22.07	–	–
DirectBW	15.65	9.87	28.45	0.58	0.261
IndirectBW	19.10	12.68	26.81	4.03	<0.001
IndirectWW	19.82	13.39	28.63	4.76	<0.001
IndirectBW vs. IndirectWW				-0.72	0.203

Notes: Mean, median, and standard deviations are presented in seconds. The difference column shows paired mean differences in task duration. p-values are from within-participant paired comparisons.

Appendix F: Assumptions for the internal validity of the within-subject analysis

In this section, I show that all three assumptions necessary for the validity of a within-subject analysis are satisfied.

Balanced panel

To assess the balanced panel assumption, I verified that all participants completed all four tasks. In the sample, every participant completed all four tasks. A McNemar test does not show differential attrition between conditions ($p = 1.0$). Therefore, I conclude that the balanced panel assumption is satisfied.

Temporal stability

Table F.1 shows whether the treatment effects in IndirectBW depend on task position by interacting the IndirectBW indicator with order dummies. In both panels, the interaction terms are jointly insignificant (Panel A: $p = 0.235$; Panel B: $p = 0.213$), indicating that order does not systematically moderate the relationship. This result supports the temporal stability of the treatment effects within the experiment.

Causal transience

Table F.2 presents sequence tests for the causal transience assumption. Sequence tests evaluate whether the within-subject treatment effect depends on which condition was seen first. Insignificant coefficients indicate no evidence of carryover, confirming that the causal transience assumption holds.

Table F.1: Temporal stability: does the IndirectBW effect vary with task position?

Sample:	Panel A *DirectBW + IndirectBW*	Panel B *IndirectBW + IndirectWW*
	(1)	(2)
IndirectBW (Order 1)	-0.061^{**}	0.108^{***}
	(0.031)	(0.030)
IndirectBW × Order 2	-0.015	-0.085^{*}
	(0.048)	(0.048)
IndirectBW × Order 3	0.038	-0.041
	(0.048)	(0.048)
IndirectBW × Order 4	-0.059	-0.084^{*}
	(0.048)	(0.047)
Joint test: interactions = 0 (p-value)	0.235	0.213
Participant fixed effects	Yes	Yes
Participants	4,327	4,327
Observations	8,654	8,654

Notes: Each regression pools observations from two tasks only. Order refers to the task position (1–4) in the full sequence. Panel A uses the DirectBW and IndirectBW sample; the dependent variable equals 1 if the Black profile is chosen. Panel B uses the IndirectBW and IndirectWW sample; the dependent variable equals 1 if the ambiguously framed option is chosen. Coefficients are from participant fixed-effects regressions with standard errors clustered by participant. The joint test reports the p-value for $\{\text{IndirectBW} \times \text{Order } k\}_{k=2,3,4} = 0$. $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.1$.

Table F.2: Sequence tests (carryover / causal transience)

Sample:	Panel A *DirectBW + IndirectBW*	Panel B *IndirectBW + IndirectWW*
	(1)	(2)
Ambiguous first	-0.022	0.006
	(0.019)	(0.020)
Participants	4,327	4,327

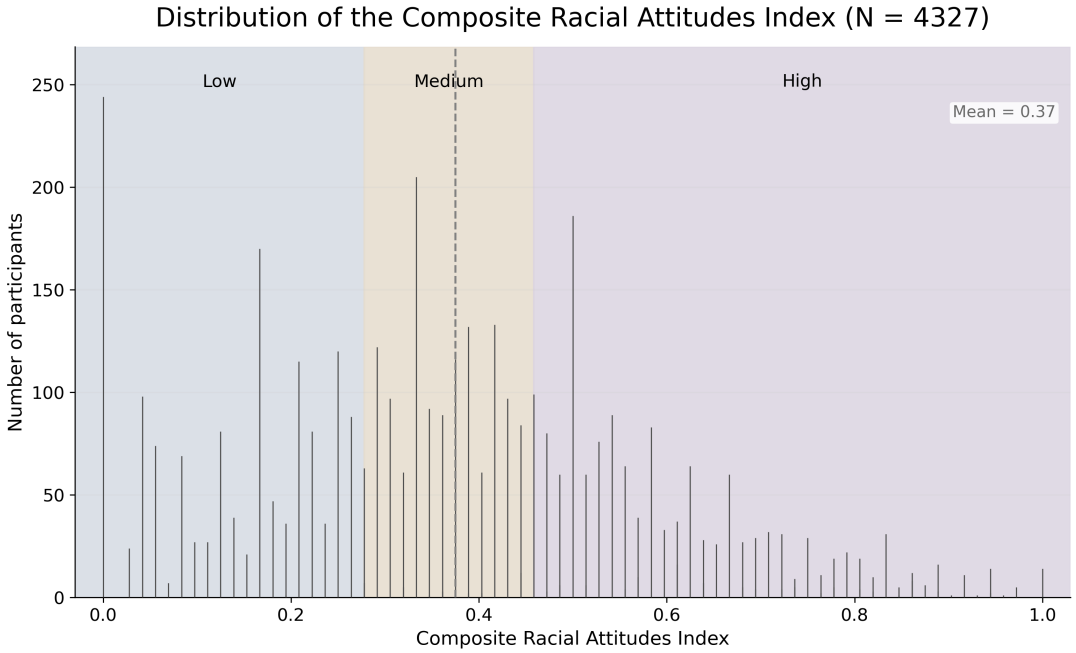
Notes: The dependent variable is the participant-level within-subject gap Δ_i: the difference in choice rates between IndirectBW and DirectBW (Panel A) and between IndirectBW and IndirectWW (Panel B). *Ambiguous first* indicates that IndirectBW appeared earlier in the sequence than the comparison condition. HC1 robust standard errors in parentheses.

Appendix G: Survey Measures

Composite Racial Attitudes Index

The Composite Racial Attitudes Index is constructed from six pre-registered attitude items toward Black Americans: two stereotype items and four symbolic-racism items. All items are coded so that higher values indicate more negative attitudes, rescaled to the unit interval because the original items are measured on different response scales, and then averaged with equal weight across items. The resulting raw index ranges from 0 to 1, with higher values indicating more negative racial attitudes toward Black Americans. For graphical presentation, I divide the raw Composite Racial Attitudes Index into Low, Medium, and High groups using tertiles.

Figure G.1: Distribution of the Composite Racial Attitudes Index



Notes: Bars show the number of participants at each value of the raw Composite Racial Attitudes Index. Shaded regions indicate the low, medium, and high composite-attitude groups used in the analysis. The dashed vertical line marks the sample mean.

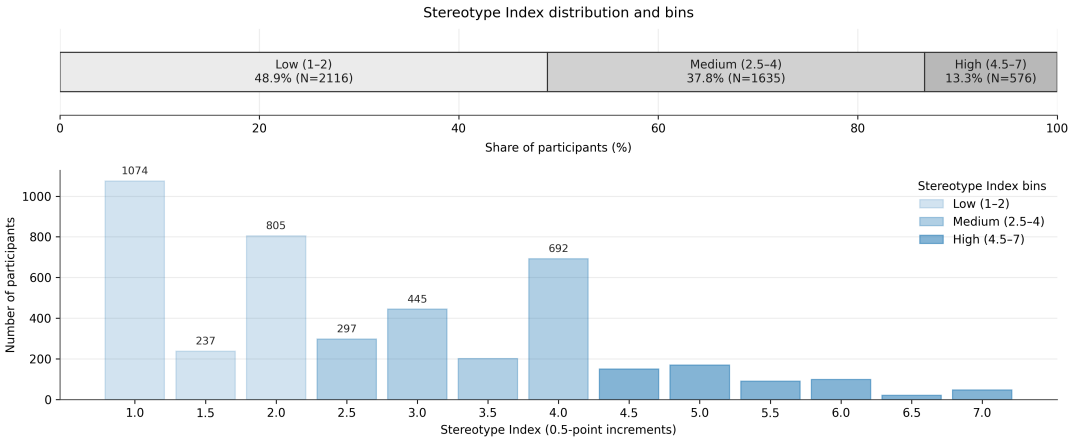
Stereotype Index

I use the two-item Stereotype Trait Ratings based on Katz and Braly (1933) to construct the Stereotype Index. I average the two items, with higher values indicating more negative stereotypes about Black Americans. The battery is as follows:

"Where would you rate Black Americans in general on this scale?"

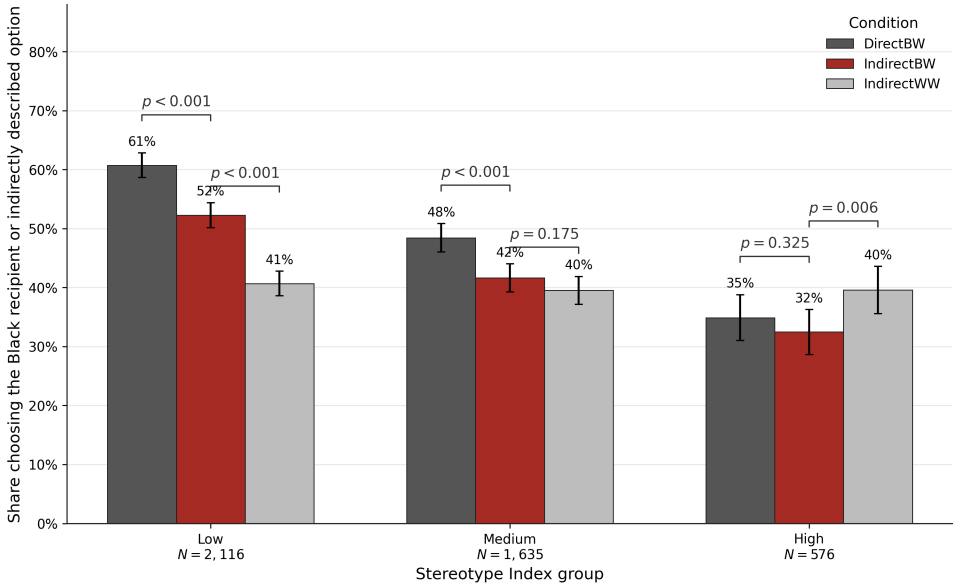
- 1 = Hard-working - 7 = Lazy
- 1 = Intelligent - 7 = Unintelligent

Figure G.2: Distribution of the Stereotype Index



Notes: The distribution of the Stereotype Index, shown in 0.5-point increments on the 1–7 scale. Histogram bars are shaded by the Low/Medium/High bins used in the heterogeneity analysis.

Figure G.3: Mean allocation choices by Stereotype Index groups: Low, Medium, High



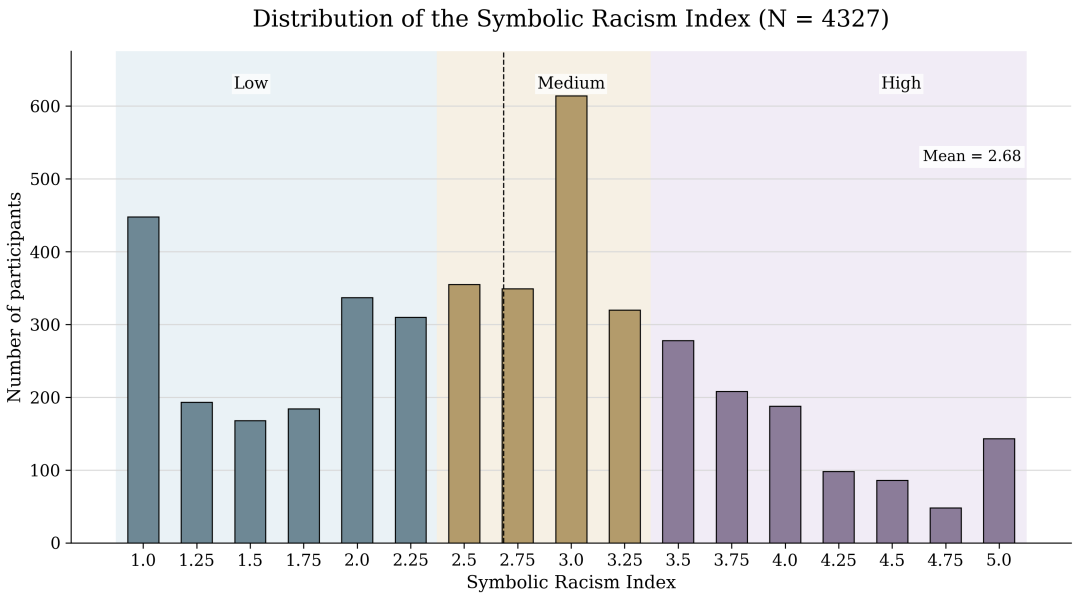
Notes: The y-axis pools two distinct outcomes: for DirectBW and IndirectBW bars, it shows the share of participants choosing the Black-recipient option; for IndirectBW and IndirectWW bars, it shows the share choosing the ambiguously framed option. Error bars are 95% confidence intervals around the mean. N denotes unique participants in each Stereotype Index group. The p-values are from paired tests within each bin.

Symbolic Racism Index

I construct the Symbolic Racism Index using the following items (Based on Henry and Sears (2002)). "Please indicate how much you agree or disagree with each statement. (1 = Strongly Disagree, 5 = Strongly Agree)"

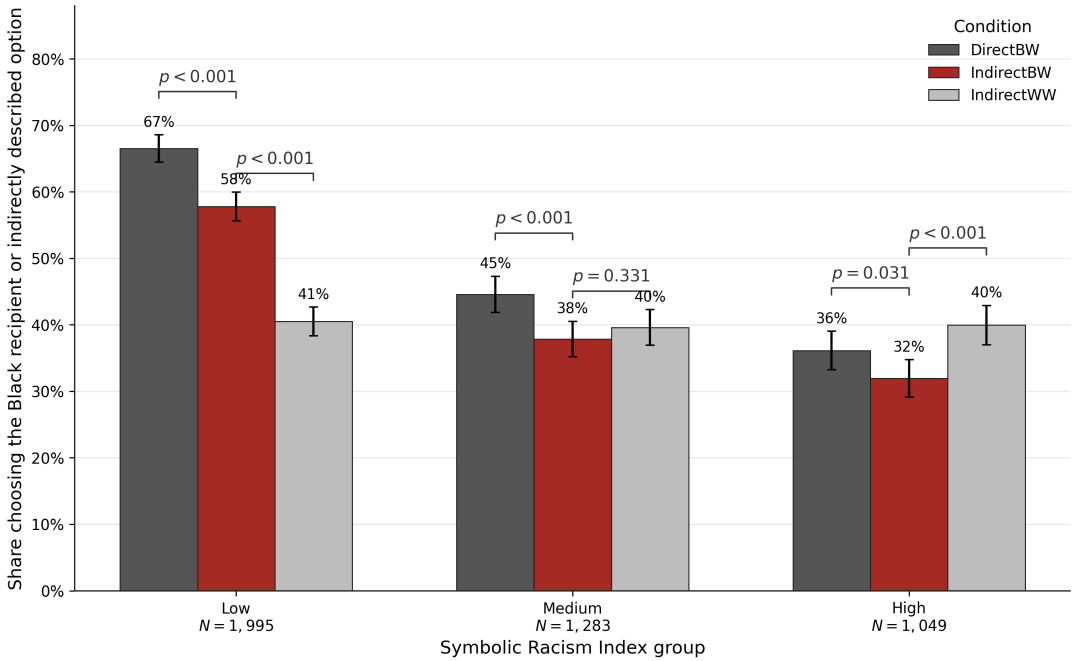
1. Irish, Italians, Jews and other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors.
2. Generations of slavery and discrimination have created conditions that make it difficult for Blacks to work their way out of the lower class. *(reverse)*
3. Over the past few years, Blacks have gotten less than they deserve. *(reverse)*
4. It's really a matter of some people not trying hard enough; if Blacks would only try harder, they could be just as well off as Whites.

Figure G.4: Distribution of the Symbolic Racism Index



Notes: Bars show the number of participants at each value of the raw Symbolic Racism Index. Shaded regions indicate the low, medium, and high symbolic racism groups used in the analysis. The dashed vertical line marks the sample mean.

Figure G.5: Mean allocation choices by the Symbolic Racism Index groups: Low, Medium, High



Notes: The figure shows mean allocation shares across the three treatment conditions for low, middle, and high Symbolic Racism Index groups. The y-axis pools two distinct outcomes: for DirectBW and IndirectBW bars, it shows the share of participants choosing the Black-recipient option; for IndirectBW and IndirectWW bars, it shows the share choosing the ambiguously framed option. Error bars show 95% confidence intervals, p-values are from within-group paired comparisons across conditions.

Table G.1: Correlations among racial-attitude measures

	Composite Index	Symbolic Racism Index	Stereotype Index
Composite Index	1.000		
Symbolic Racism Index	0.937***	1.000	
Stereotype Index	0.674***	0.372***	1.000

Notes: The table shows participant-level Pearson correlations among the Composite Racial Attitudes Index, the Symbolic Racism Index, and the Stereotype Index. All measures are coded so that higher values indicate more negative attitudes toward Black Americans. Composite Index refers to the Composite Racial Attitudes Index. $^{***}p < 0.01$.

Table G.2: Correlations among the components of the Composite Racial Attitudes Index

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Hardworking vs. lazy	1.000					
(2) Intelligent vs. unintelligent	0.869***	1.000				
(3) Overcome prejudice without favors	0.330***	0.288***	1.000			
(4) Slavery made it hard to get ahead (rev.)	0.287***	0.255***	0.556***	1.000		
(5) Get less than they deserve (rev.)	0.333***	0.292***	0.557***	0.712***	1.000	
(6) Should try harder	0.329***	0.291***	0.763***	0.524***	0.486***	1.000

Notes: The table shows Pearson correlations among the six items used to construct the Composite Racial Attitudes Index. Items (1) and (2) are stereotype ratings of Black Americans on 1–7 scales: (1) asks respondents to rate Black Americans between *hardworking* and *lazy*, and (2) asks respondents to rate Black Americans between *intelligent* and *unintelligent*. Items (3)–(6) are symbolic-racism items measured on 1–5 agreement scales: (3) "Irish, Italians, Jewish and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors"; (4) "Generations of slavery and discrimination have created conditions that make it difficult for Blacks to work their way out of the lower class"; (5) "On average, Black Americans get less than they deserve"; and (6) "It is really a matter of some people not trying hard enough; if Blacks would only try harder, they could be just as well off as whites." Items (4) and (5) are reverse-coded before index construction so that, for all six items, higher values indicate more negative attitudes toward Black Americans. Spearman correlations are very similar. $^{***}p < 0.01$.

Appendix H: Robustness to the gender of the recipient profile

As an additional robustness check, I examine whether the results differ across the male-profile and female-profile versions of the tasks. In the survey, each participant was assigned to one of these two versions and then completed the four experimental tasks within that assigned gender arm.

Table H.1 shows that the qualitative pattern of the main results is the same in both arms. In the DirectBW–IndirectBW comparison, attaching ambiguous framing to the Black profile reduces Black-recipient choice in both the male-profile arm (-0.0512, $p < 0.01$) and the female-profile arm (-0.0933, $p < 0.01$). The reduction is somewhat larger in the female-profile arm (-0.0933 vs. -0.0512), and the pooled interaction test rejects equality of treatment effects across arms (-0.0446, $p = 0.0196$). In line with the main results of the paper, in the IndirectWW–IndirectBW comparison, the effect is positive in both arms (0.0779, $p < 0.01$, in the male-profile arm; 0.0650, $p < 0.01$, in the female-profile arm), and the difference across arms is small and statistically insignificant (-0.0110, $p = 0.5630$). Thus, the main conclusion remains unchanged: even when the sample is divided by the gender of the recipient, participants are not less willing to choose the ambiguously framed option when it benefits the Black recipient.

Table H.2 further shows that the heterogeneity pattern by the Composite Racial Attitudes Index is directionally similar across the two gender arms. In both arms, participants in the low-attitudes group are more likely to choose the ambiguous option when it benefits the Black recipient (0.2575, $p < 0.01$, in the male-profile arm; 0.1999, $p < 0.01$, in the female-profile arm), whereas participants in the high-attitudes group are less likely to do so (-0.0564, $p < 0.05$, in the male-profile arm; -0.0576, $p < 0.05$, in the female-profile arm). Taken together, these results suggest that the recipient's gender does not alter the main qualitative conclusions.

Table H.1: Effects of race and ambiguous framing on allocation decisions by recipient-profile gender

Sample:	*DirectBW + IndirectBW*		*IndirectBW + IndirectWW*	
	Male-profile arm	Female-profile arm	Male-profile arm	Female-profile arm
	(1)	(2)	(3)	(4)
IndirectBW	-0.0512***	-0.0933***	0.0779***	0.0650***
	(0.0130)	(0.0139)	(0.0142)	(0.0148)
Control mean	0.5258	0.5272	0.4223	0.3798
Participant fixed effects	Yes	Yes	Yes	Yes
Participants	2,155	2,172	2,155	2,172
Observations	4,310	4,344	4,310	4,344
Female − Male TE	−0.0446**		−0.0110	
Interaction p-value	0.0196		0.5630	

Notes: Each regression pools observations from two tasks only. The dependent variable equals 1 if the Black-recipient option is selected in the DirectBW/IndirectBW comparison (columns (1)–(2)), and 1 if the ambiguously framed option is selected in the IndirectBW/IndirectWW comparison (columns (3)–(4)), where the ambiguous option is attached to the Black recipient in IndirectBW and to the White recipient in IndirectWW. Columns (1) and (3) use the male-profile version of the task battery; columns (2) and (4) use the female-profile version. "Control mean" is the mean of the dependent variable in the control condition (DirectBW in columns (1)–(2); IndirectWW in columns (3)–(4)). All specifications include participant fixed effects and standard errors clustered by participant. Controls match the main specification. The interaction rows report pooled fixed-effects estimates testing whether the treatment effect differs across the male- and female-profile versions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table H.2: Effects of race and ambiguous framing on allocation decisions by recipient-profile gender and Composite Racial Attitudes Index group

Sample:	*DirectBW + IndirectBW*		*IndirectBW + IndirectWW*	
Composite Index group	Male-profile arm	Female-profile arm	Male-profile arm	Female-profile arm
	(1)	(2)	(3)	(4)
Low	-0.0803***	-0.1392***	0.2575***	0.1999***
	(0.0219)	(0.0239)	(0.0249)	(0.0261)
Medium	-0.0296	-0.0793***	0.0329	0.0573**
	(0.0240)	(0.0256)	(0.0241)	(0.0258)
High	-0.0440**	-0.0581**	-0.0564**	-0.0576**
	(0.0218)	(0.0229)	(0.0232)	(0.0237)

Notes: Each regression pools observations from two tasks only. The dependent variable equals 1 if the Black-recipient option is selected in the DirectBW/IndirectBW comparison (columns (1)–(2)), and 1 if the ambiguously framed option is selected in the IndirectBW/IndirectWW comparison (columns (3)–(4)), where the ambiguous option is attached to the Black recipient in IndirectBW and to the White recipient in IndirectWW. Rows correspond to within-arm tertiles of the Composite Racial Attitudes Index. All specifications include participant fixed effects and standard errors clustered by participant, and use the same choice-level controls as in the main specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix I: Experimental Instructions

The following describes the experimental instructions of the main experiment (Stage 2). Block names were not displayed to participants.

Block 1: Consent Form

Greeting

> Thank you for participating in this survey!
>
> It is important for us that all your answers are as accurate and truthful as possible.
>
> On the next page, you will see the Consent form. Please read it carefully before proceeding.

[page break]

Consent form

> Before we begin, please read the following carefully. By participating, you will provide demographic information (e.g., age, gender, racial/ethnic origin, religious/political beliefs etc.) and complete tasks involving allocation decisions.
>
> You can participate in this survey once. Your responses will be anonymized, and no identifying information will be linked to your data. The data will be used only for research purposes. You may withdraw at any time, and your data will be deleted upon request. Compensation is provided only upon full completion of the study. It is important that you pay close attention and answer all questions as accurately and truthfully as possible.
>
> If you are 18 years of age and older, agree with the statements above, and freely consent to participate in the study, please click on the "I Agree" button to start.

Response options: {I Agree; I Disagree}.

Block 2: Prolific ID

What is your Prolific ID:

Please note that this response should auto-fill with the correct ID.

[text box]

Block 3: Screener Validation

What is your race?

Response options: {White; Black or African American; Asian; Hispanic or Latino; Other *[text box]*}.

Block 4: Demographics 1

What is your age? *[text box]*

What is your gender?

Response options: {Male; Female; Other *[text box]*}.

What is your religious affiliation?

Response options: {Christian; Muslim; Hindu; Jewish; Buddhist; Not Religious; Other *[text box]*}.

What are your political views?

Response options: {Very liberal; Liberal; Moderate; Conservative; Very conservative; Apolitical}.

[page break]

What is your current state of residence?

Please select your state from the list. *[drop-down list of U.S. states]*

What is your current employment status?

Response options: {Employed full-time; Employed part-time; Unemployed; Retired; Student; Self-employed; Homemaker}.

What is your highest level of education?

Response options: {Less than high school; High school diploma or equivalent; Some college; Associate degree; Bachelor's degree; Master's degree; Doctoral degree}.

What is your sexual orientation?

Response options: {Heterosexual (Straight); Homosexual (Gay/Lesbian); Bisexual; Other *[text box]*}.

[page break]

To ensure data quality, we occasionally include a question to check attentiveness. For this question only, please select "Somewhat Disagree" regardless of your actual opinion.

Response options: {Strongly Disagree; Somewhat Disagree; Neither Agree nor Disagree; Somewhat Agree; Strongly Agree}.

[page break]

What is your household income?

Response options: {Less than $10,000; $10,000 to $14,999; $15,000 to $24,999; $25,000 to $34,999; $35,000 to $49,999; $50,000 to $74,999; $75,000 to $99,999; $100,000 to $149,999; $150,000 to $199,999; $200,000 or more}.

What is your marital status?

Response options: {Single, never married; Married; Widowed; Divorced; Separated; In a civil union}.

How many children do you have?

Response options: {0; 1; 2; 3; 4 and more}.

The questions in this block were presented individually on separate pages.

Block 5: Tasks Intro

Next, you will complete a series of tasks. The profiles in the tasks are real. After all participants complete the survey, some participants will be randomly selected, and one of their choices will be randomly chosen to be implemented.

In each task, you will see two profiles labeled Option A and Option B. Please carefully review both options and select the one you prefer. Each of your choices may have a real impact on payments to the other individual. Therefore, please pay attention and make all your choices carefully.

Block 6: Allocation Tasks

In the main study, each participant was assigned to either the male-profile version or the female-profile version of the task battery. Within the assigned version, the participant completed the following four conditions, shown in randomized order. Within each condition, the profiles were randomized across the left and right positions.

DirectBW. Participants chose between a Black profile and a White profile. Both options were described as: "This profile receives 100 cents." Response options: {Option A; Option B}.

DirectWW. Participants chose between two White profiles. Both options were described as: "This profile receives 100 cents." Response options: {Option A; Option B}.

IndirectBW. Participants chose between a Black profile and a White profile. One option was described as "This profile receives the larger amount from the following two: either 100 or 50 cents," while the other option was described as "This profile receives 100 cents." Response options: {Option A; Option B}.

IndirectWW. Participants chose between two White profiles. One option was described as "This profile receives the larger amount from the following two: either 100 or 50 cents," while the other option was described as "This profile receives 100 cents." Response options: {Option A; Option B}.

Block 7: Clarity

Were the instructions for the following task clear to you?

"This profile receives the larger amount from the following two: either 100 or 50 cents"

Response options: {Yes, I understood the task clearly.; No, the instructions were not clear.; Other *[text box]*}.

Block 8: Demographics 2 and Preference Measures

What is your favorite sport?

Response options: {Soccer; Basketball; Tennis; Baseball; Football; Swimming; Running}.

[page break]

What is your favorite color?

Please select from the list. *[drop-down list]* Response options: {Blue; Green; Pink; Black; Purple; Red; Yellow; Gray.}.

[page break]

Do you prefer coffee or tea?

Response options: {Coffee; Tea}.

[page break]

On a scale from 0 to 10, how willing are you to take risks in general?

0 = Completely risk-averse (avoid risk at all costs)

10 = Very risk-seeking (comfortable taking high risks)

[slider from 0 to 10]

Please use the slider to indicate your risk preference.

[page break]

The following questions ask about your decision-making preferences. Please indicate how much you agree or disagree with each of the following statements.

Response scale: {Strongly Disagree; Disagree; Neither Agree nor Disagree; Agree; Strongly Agree}.

Statements: (1) I prefer to make decisions when the chances of each outcome are known. (2) I feel uneasy when I have to make a decision without knowing all the probabilities involved. (3) I tend to avoid options when I do not know how likely the outcomes are. (4)

I am comfortable making decisions when the risks are clear, even if they involve losing something. (5) I prefer problems where all the facts are clearly laid out.

Block 9: Racial Attitudes Battery

> Thank you for completing the previous tasks. In this short section, we would like your general views on various topics. Your responses are confidential and important for our research.
>
> (7-point scale; select one number for each question)

> **Where would you rate Black Americans in general on this scale?**
>
> 1 (Hard-working), 2, 3, 4 (Neither), 5, 6, 7 (Lazy)

[7-point response scale]

> **Where would you rate Black Americans in general on this scale?**
>
> 1 (Intelligent), 2, 3, 4 (Neither), 5, 6, 7 (Unintelligent)

[7-point response scale]

[page break]

> **Please indicate how much you agree or disagree with each of the following statements.**

Response scale: {Strongly Disagree; Disagree; Neither Agree nor Disagree; Agree; Strongly Agree}.

Statements: (1) I attempt to act in non-prejudiced ways toward Black people because it is personally important to me. (2) I try to act non-prejudiced toward Black people because of my personal values. (3) Being non-prejudiced toward Black people is important to my self-concept.

[page break]

> **Please indicate how much you agree or disagree with each of the following statements.**

Response scale: {Strongly Disagree; Disagree; Neither Agree nor Disagree; Agree; Strongly Agree}.

Statements: (1) Irish, Italians, Jews and other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors. (2) Generations of slavery and discrimination have created conditions that make it difficult for Blacks to work their way out of the lower class. (3) Over the past few years, Blacks have gotten less than they deserve. (4) It is really a matter of some people not trying hard enough; if Blacks would only try harder, they could be just as well off as Whites.

End-of-survey message

> Thank you for taking part in this study. Please click the button below to be redirected back to Prolific and register your submission.

References

Dennis J Aigner and Glen G Cain. Statistical theories of discrimination in labor markets. *Ilr Review*, 30(2):175–187, 1977.

David Arnold, Will Dobbie, and Crystal S Yang. Racial bias in bail decisions. *The Quarterly Journal of Economics*, 133(4):1885–1932, 2018.

Kenneth J. Arrow. The theory of discrimination. In Orley Ashenfelter and Albert Rees, editors, *Discrimination in Labor Markets*, pages 3–33. Princeton University Press, Princeton, NJ, 1973.

Kenneth J. Arrow. What has economics to say about racial discrimination? *Journal of Economic Perspectives*, 12(2):91–100, 1998. doi: 10.1257/jep.12.2.91.

Philipp Bach, Victor Chernozhukov, Malte S. Kurz, and Martin Spindler. DoubleML—an object-oriented implementation of double machine learning in Python. *Journal of Machine Learning Research*, 23(53):1–6, 2022. URL `https://www.jmlr.org/papers/v23/21-0862.html`.

Kai Barron, Ruth Ditlmann, Stefan Gehrig, and Sebastian Schweighofer-Kodritsch. Explicit and implicit belief-based gender discrimination: A hiring experiment. *Management Science*, 71(2):1600–1622, 2025. doi: 10.1287/mnsc.2022.01229.

Vojtěch Bartoš, Ulrich Glogowsky, and Johannes Rincke. The color of knowledge: Impacts of tutor race on learning and performance. Discussion Paper 20945, CEPR, 2025. URL `https://cepr.org/publications/dp20945`.

Michal Bauer and Julie Chytilová. Ethnic hostility and conflict. CEPR Discussion Paper DP20402, CEPR, July 2025. URL `https://cepr.org/publications/dp20402`.

Michal Bauer, Jana Cahlíková, Julie Chytilová, and Tomáš Želinský. Social contagion of ethnic hostility. *Proceedings of the National Academy of Sciences*, 115(19):4881–4886, 2018. doi: 10.1073/pnas.1720317115.

Michal Bauer, Jana Cahlíková, Julie Chytilová, Gérard Roland, and Tomáš Želinský. Shifting punishment on minorities: Experimental evidence of scapegoating. *The Economic Journal*, 133(652):1626–1640, 2023. doi: 10.1093/ej/uead005.

Michal Bauer, Jana Cahlíková, Dagmara Celik Katreniak, Julie Chytilová, Lubomír Cingl, and Tomáš Želinský. Nastiness in groups. *Journal of the European Economic Association*, 22(5):2075–2107, 2024. doi: 10.1093/jeea/jvad072.

Gary S. Becker. *The Economics of Discrimination*. University of Chicago Press, Chicago, 1957.

Roland Bénabou and Jean Tirole. Incentives and prosocial behavior. *American Economic Review*, 96(5):1652–1678, 2006. doi: 10.1257/aer.96.5.1652.

Marianne Bertrand and Esther Duflo. Field experiments on discrimination. In Abhijit Vinayak Banerjee and Esther Duflo, editors, *Handbook of Economic Field Experiments*, volume 1, pages 309–393. Elsevier, 2017. doi: 10.1016/bs.hefe.2016.08.004.

Marianne Bertrand and Sendhil Mullainathan. Are emily and greg more employable than lakisha and jamal? a field experiment on labor market discrimination. *American Economic Review*, 94(4):991–1013, 2004. doi: 10.1257/0002828042002561.

J. Aislinn Bohren, Kareem Haggag, Alex Imas, and Devin G. Pope. Inaccurate statistical discrimination: An identification problem. *The Review of Economics and Statistics*, 107 (3):605–620, 2025. doi: 10.1162/rest_a_01367.

Leonardo Bursztyn, Georgy Egorov, and Stefano Fiorin. From extreme to mainstream: The erosion of social norms. *American Economic Review*, 110(11):3522–3548, 2020. doi: 10.1257/aer.20171175.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097.

Jason Dana, Roberto A. Weber, and Jason Xi Kuang. Exploiting moral wiggle room: Experiments demonstrating an illusory preference for fairness. *Economic Theory*, 33(1): 67–80, 2007. doi: 10.1007/s00199-006-0153-z.

Ellora Derenoncourt, Chi Hyun Kim, Moritz Kuhn, and Moritz Schularick. Wealth of two nations: The US racial wealth gap, 1860–2020. *The Quarterly Journal of Economics*, 139(2):693–750, 2024.

Sanjaya DeSilva and Yuval Elmelech. Housing inequality in the united states: Explaining the white-minority disparities in homeownership. *Housing Studies*, 27(1):1–26, 2012.

Samuel P. Douglas, Phillip J. Ewell, and Markus Brauer. Data quality in online experiments: Comparing platforms and recruitment methods. *PLOS ONE*, 18(5):e0284998, 2023. doi: 10.1371/journal.pone.0284998.

Benjamin Edelman, Michael Luca, and Dan Svirsky. Racial discrimination in the sharing economy: Evidence from a field experiment. *American Economic Journal: Applied Economics*, 9(2):1–22, 2017. doi: 10.1257/app.20160213.

Griffin Sims Edwards and Stephen Rushin. The Effect of President Trump's Election on Hate Crimes. *SSRN Electronic Journal*, 2018. doi: 10.2139/ssrn.3102652. URL https://ssrn.com/abstract=3102652.

Daniel Ellsberg. Risk, ambiguity, and the savage axioms. *The Quarterly Journal of Economics*, 75(4):643–669, 1961.

Benjamin Enke, Ricardo Rodriguez-Padilla, and Florian Zimmermann. Moral universalism: Measurement and economic relevance. *Management Science*, 2021.

Nicholas Epley and Thomas Gilovich. The mechanics of motivated reasoning. *Journal of Economic Perspectives*, 30(3):133–140, 2016.

Christine L Exley. Excusing selfishness in charitable giving: The role of risk. *The Review of Economic Studies*, 83(2):587–628, 2016.

Christine L Exley and Judd B Kessler. Motivated errors. *American Economic Review*, 114(4):961–987, 2024.

Markus Eyting. Why do we discriminate? Directional beliefs and biased information processing. *Management Science*, 2026. doi: 10.1287/mnsc.2025.01818. URL https://doi.org/10.1287/mnsc.2025.01818. Articles in Advance.

Chaim Fershtman and Uri Gneezy. Discrimination in a segmented society: An experimental approach. *The Quarterly Journal of Economics*, 116(1):351–377, 2001. doi: 10.1162/003355301556338.

Pauline Grosjean, Federico Masera, and Hasin Yousaf. Inflammatory political campaigns and racial bias in policing. *The Quarterly Journal of Economics*, 138(1):413–463, 2023. doi: 10.1093/qje/qjac037.

Zachary Grossman and Joël J. van der Weele. Self-image and willful ignorance in social decisions. *Journal of the European Economic Association*, 15(1):173–217, 2017. doi: 10.1093/jeea/jvw001.

Shreyas Gupta, Thomas Rigotti, and Alistair Wilson. The experimenters' dilemma: Inferential preferences over populations. *Experimental Economics*, 24:1442–1474, 2021. doi: 10.1007/s10683-021-09714-4.

Ingar Haaland and Christopher Roth. Beliefs about Racial Discrimination and Support for Pro-Black Policies. *The Review of Economics and Statistics*, 105(1):40–53, 2023. doi: 10.1162/rest_a_01036.

Emily C. Haisley and Roberto A. Weber. Self-serving interpretations of ambiguity in other-regarding behavior. *Games and Economic Behavior*, 68(2):614–625, 2010. doi: 10.1016/j.geb.2009.08.002.

Patrick J Henry and David O Sears. The symbolic racism 2000 scale. *Political Psychology*, 23(2):253–283, 2002.

Daniel Katz and Kenneth Braly. Racial stereotypes of one hundred college students. *The Journal of Abnormal and Social Psychology*, 28(3):280, 1933.

Ziva Kunda. The case for motivated reasoning. *Psychological Bulletin*, 108(3):480, 1990.

Tom Lane. Discrimination in the laboratory: A meta-analysis of economics experiments. *European Economic Review*, 90:375–402, 2016. doi: 10.1016/j.euroecorev.2015.11.011.

Liang Li and Tom Greene. A weighting analogue to pair matching in propensity score analysis. *International Journal of Biostatistics*, 9(2):215–234, 2013. doi: 10.1515/ijb-2012-0030.

John A. List. The nature and extent of discrimination in the marketplace: Evidence from the field. *The Quarterly Journal of Economics*, 119(1):49–89, 2004. doi: 10.1162/003355304772839588.

Benjamin J. Newman, Jennifer L. Merolla, Sono Shah, Danielle Casarez Lemi, Loren Collingwood, and S. Karthick Ramakrishnan. The Trump Effect: An Experimental Investigation of the Emboldening Effect of Racially Inflammatory Elite Communication. *British Journal of Political Science*, 51(3):1138–1159, 2021. doi: 10.1017/S0007123419000590.

Brian A. Nosek, Frederick L. Smyth, Jeffrey J. Hansen, Thierry Devos, Nicole M. Lindner, Kate A. Ranganath, Colin Tucker Smith, Kristina R. Olson, Dolly Chugh, Anthony G. Greenwald, and Mahzarin R. Banaji. Pervasiveness and correlates of implicit attitudes and stereotypes. *European Review of Social Psychology*, 18(1):36–88, 2007. doi: 10.1080/10463280701489053.

Brian A. Nosek, Mahzarin R. Banaji, and John T. Jost. The politics of intergroup attitudes. In John T. Jost, Aaron C. Kay, and Hulda Thorisdottir, editors, *Social and Psychological Bases of Ideology and System Justification*, pages 480–506. Oxford University Press, 2009. doi: 10.1093/acprof:oso/9780195320916.003.020.

Stefan Palan and Christian Schitter. Prolific.ac—a subject pool for online experiments. *Journal of Behavioral and Experimental Finance*, 17:22–27, 2018. doi: 10.1016/j.jbef.2017.12.004.

Eyal Peer, David Rothschild, Andrew Gordon, Zak Evernden, and Ekaterina Damer. Data quality of platforms and panels for online behavioral research. *Behavior Research Methods*, 54(4):1643–1662, 2022.

Edmund S. Phelps. The statistical theory of racism and sexism. *American Economic Review*, 62(4):659–661, September 1972.

Kai Spiekermann and Arne Weiss. Objective and subjective compliance: A norm-based explanation of "moral wiggle room". *Games and Economic Behavior*, 96:170–183, 2016. doi: 10.1016/j.geb.2015.11.007.

Kim Uittenhove, Sophie Jeanneret, and Evie Vergauwe. Online samples in cognitive psychology: Prolific vs MTurk vs university students. *Behavior Research Methods*, 54 (6):3216–3234, 2022. doi: 10.3758/s13428-021-01777-1.

U.S. Bureau of Labor Statistics. Usual weekly earnings of wage and salary workers: Third quarter 2025. News Release USDL-25-1529, U.S. Bureau of Labor Statistics, December 2025. URL `https://www.bls.gov/news.release/pdf/wkyeng.pdf`. Accessed 2026-01-22.